Wave optics

light :- 1675 : Newton's corpuscular theory

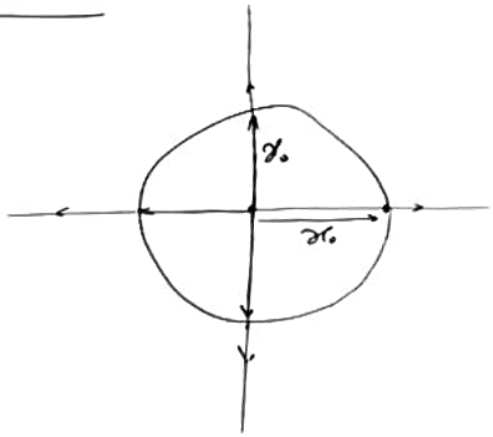
- light is particle known as corpuscle.
  - light, small and elastic
- in denser medium corpuscle travels slowly, and in rarer medium it travels faster

Huygen wave theory {1678}

- light is longitudinal wave (X)
- light is mechanical wave (X)
- **wavefront** :- if it is an imaginary surface, at which each point has equal **phase**.

Wave optics

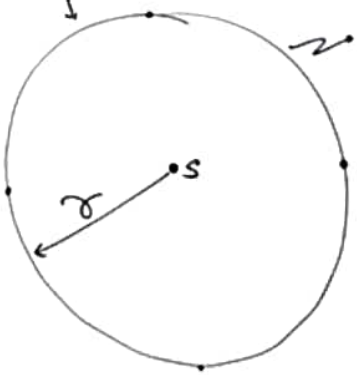
Point source



$y = A \sin(\omega t - kx)$

$\omega t - kx_0$

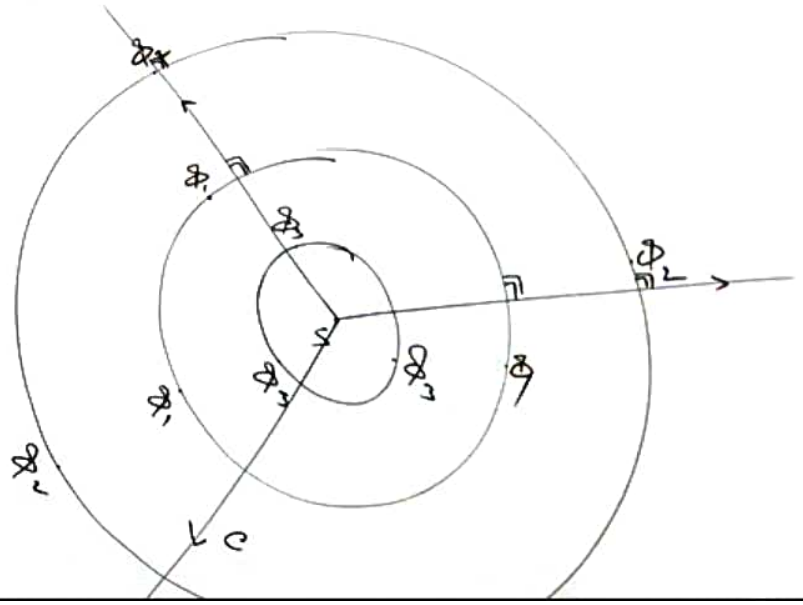
imaginary sphere



Wave optics

Point source

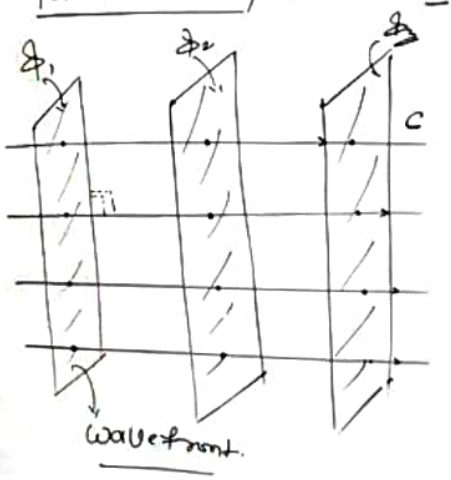
- wavefront is imaginary sphere having centre at source itself
- dirn of propagation of wave is always normal to wavefront.
- dirn of velocity of wave is always ⊥ to wavefront.



competishun

Wave optics

Parallel Beam / Source at ∞.

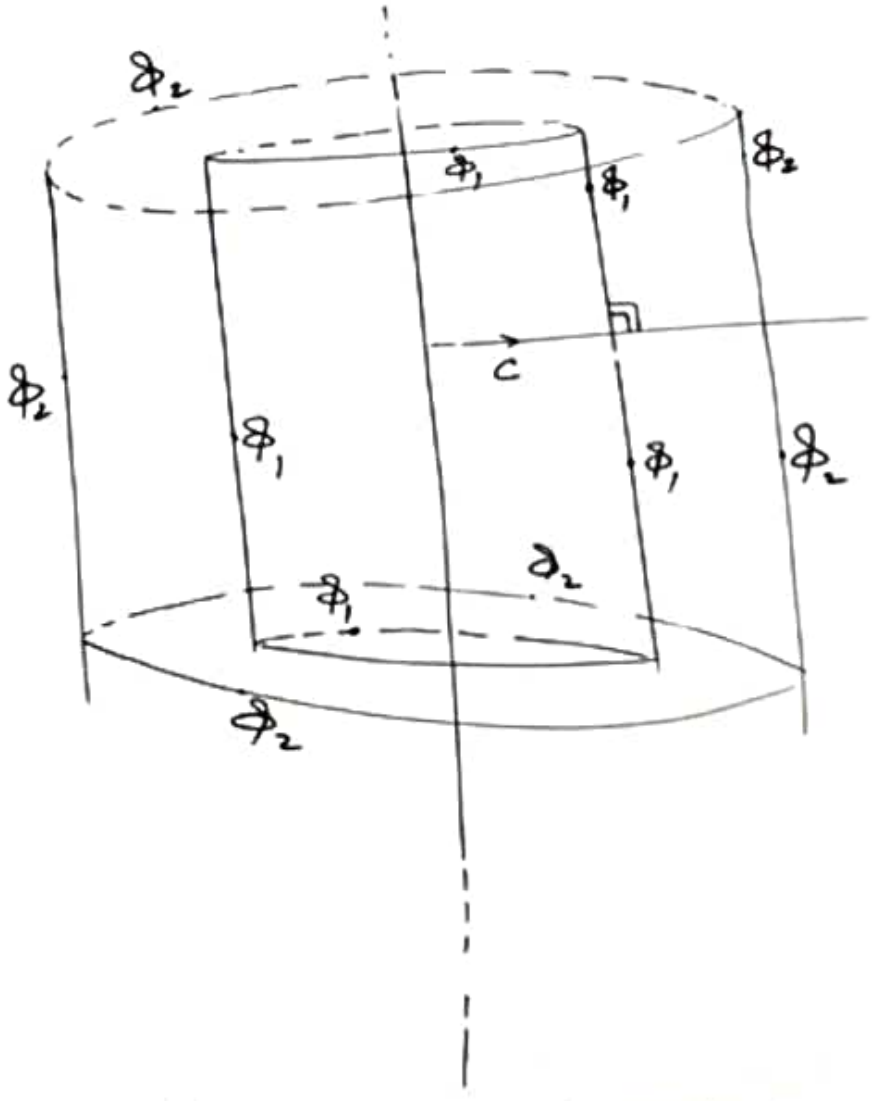


→ Each wavefront: Plane

Wave optics

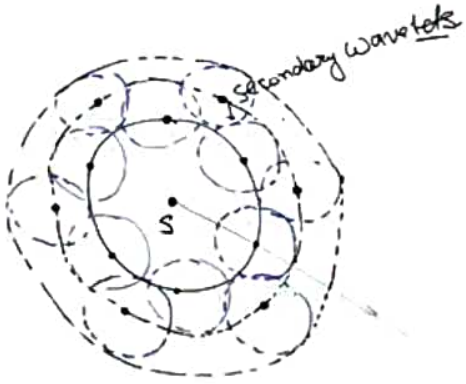
Infinitely long light source / line source

wavefront will be imaginary cylinders having axis as line source itself



Wave optics

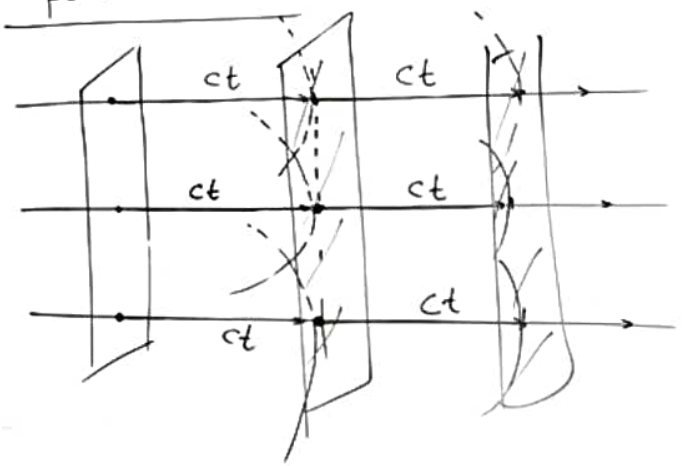
Huygen principle



Wave optics

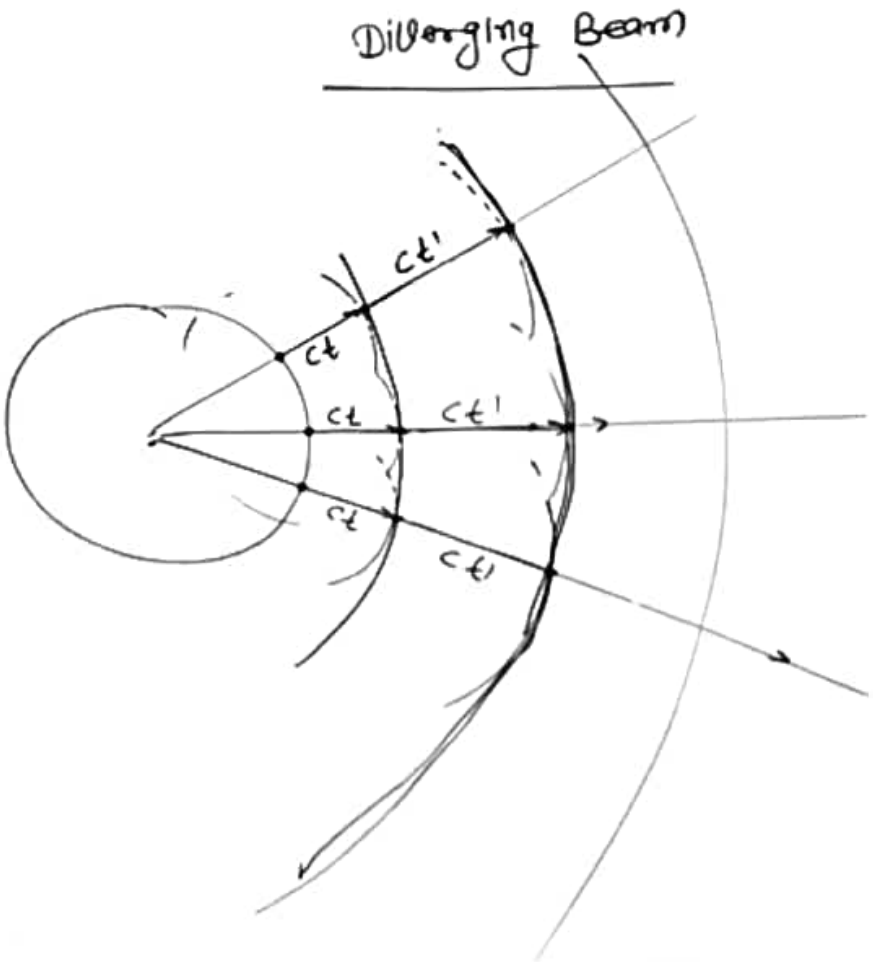
Huygen principle

parallel Beam



Wave optics

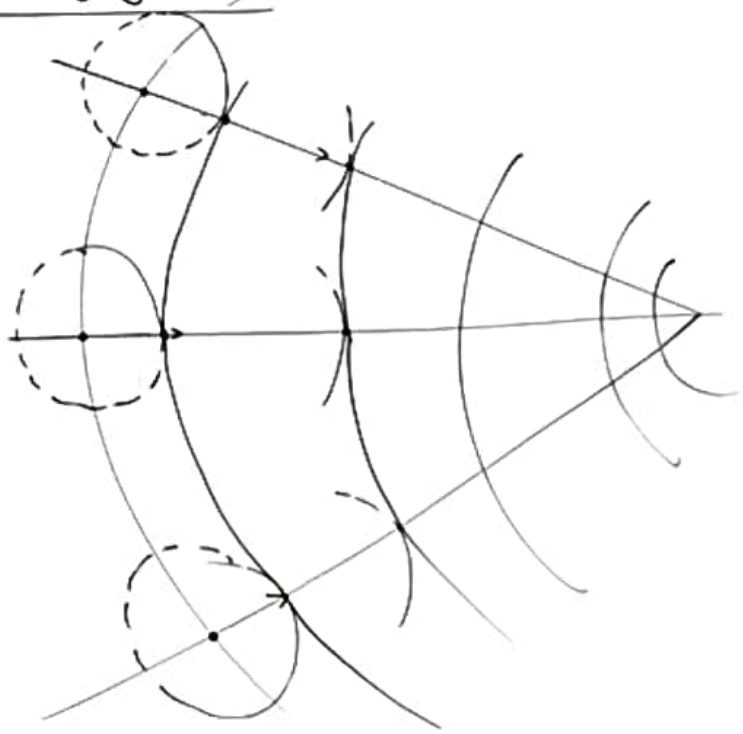
Diverging Beam



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Wave optics

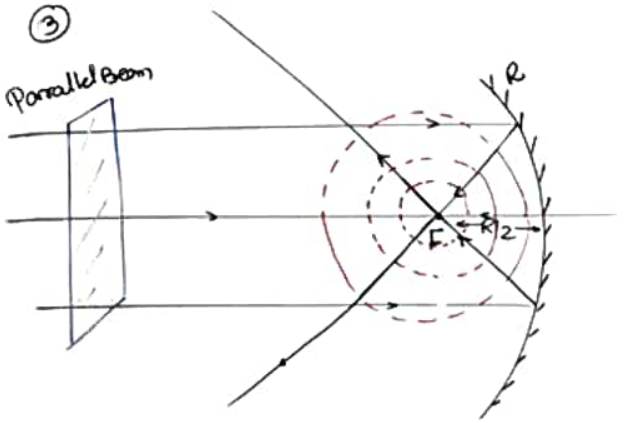
Converging Beam



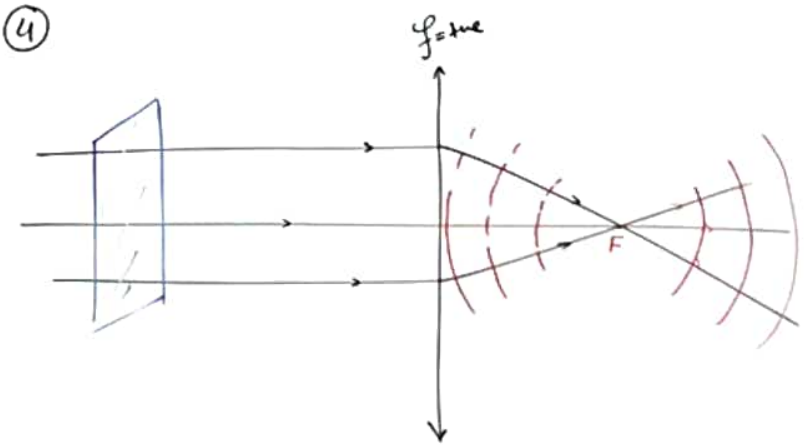
Wave optics

Draw wavefronts of these ray diagrams.

③



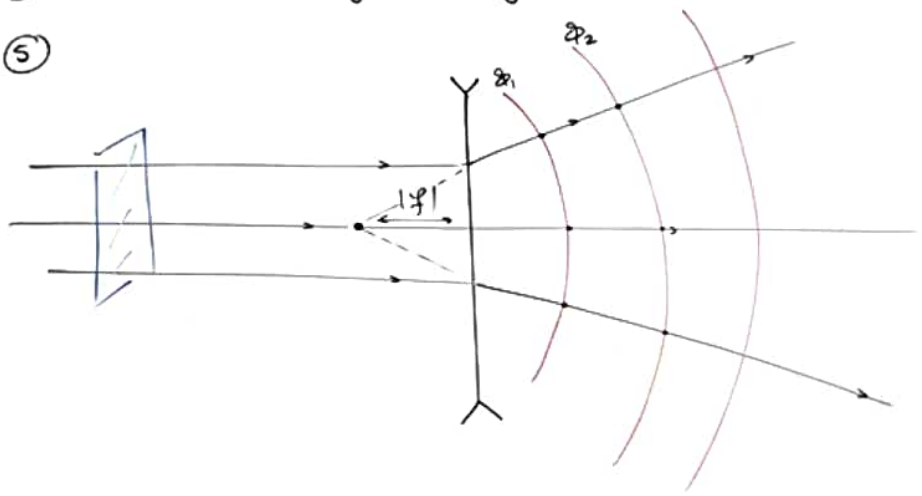
④



Wave optics

Draw wavefronts of these ray diagrams.

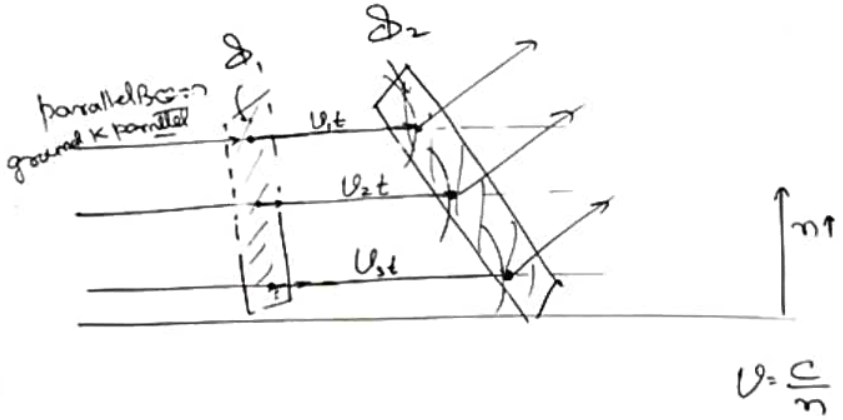
(5)



time taken to travel from 1 wavefront to another along different paths will be always equal.

Wave optics

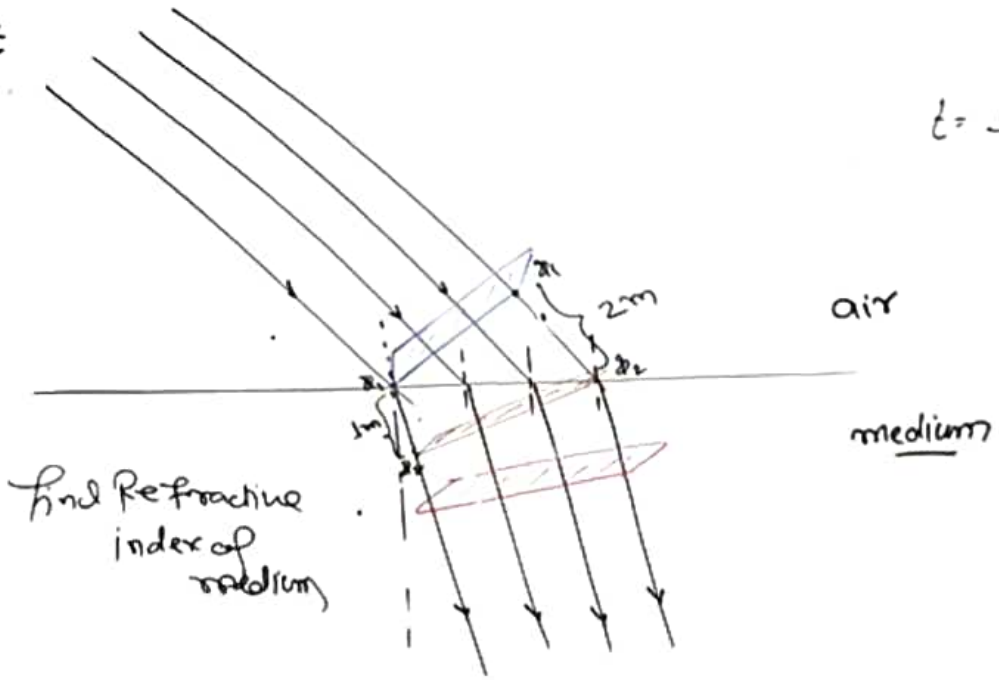
Ex



$v = \frac{c}{n}$

Wave optics

Ex



$t = \frac{2}{c} = \frac{1}{c/n}$

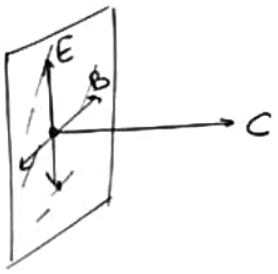
$n = 2$ Ans.

Wave optics

Maxwell's EM theory [1873]

- non-mechanical wave
- transverse wave
- light - combination of time varying electric and magnetic field

$\vec{E} \perp \vec{B} \perp \vec{c} \perp \vec{E}$



Superposition of waves

$E_1 = E_{01}\sin(\omega t)$

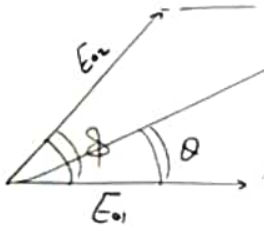
$E_2 = E_{02}\sin(\omega t+\phi)$

phase difference = ϕ

$E_{Res} = E_1 + E_2$

$= E_{01}\sin\omega t + E_{02}\sin(\omega t+\phi)$

$E_{Res} = \sqrt{E_{01}^2 + E_{02}^2 + 2E_{01}E_{02}\cos\phi} = E_0$



$\tan\theta = \dfrac{E_{02}\sin\phi}{E_{01}+E_{02}\cos\phi}$

$E_{Res} = E_0\sin(\omega t + \theta)$

Superposition of waves

$E_1 = E_{01} \sin(\omega t) \rightarrow I_1$

$E_2 = E_{02} \sin(\omega t + \phi) \rightarrow I_2$

phase difference = ϕ

$I \propto E_0^2$

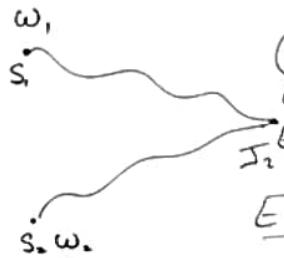
$(E_{Res})_{amp} = \sqrt{E_{01}^2 + E_{02}^2 + 2E_{01}E_{02}\cos\phi}$ Rem!

$(E_{Res})_{max}^2 = E_{01}^2 + E_{02}^2 + 2E_{01}E_{02}\cos\phi$

$I_{Res} = I_1 + I_2 + 2\sqrt{I_1}\sqrt{I_2}\cos\phi$ Rem!

Interference of incoherent waves

different ω



I_1

$E_1 = E_{01} \sin(\omega_1 t + \theta_1)$

I_2 $E_2 = E_{02} \sin(\omega_2 t + \theta_2)$

$(\omega_2 t + \theta_2) - (\omega_1 t + \theta_1)$

$E_{Res} = \sqrt{E_{01}^2 + E_{02}^2 + 2E_{01}E_{02}\cos((\omega_2 - \omega_1)t + (\theta_2 - \theta_1))}$

$\sqrt{E_{\omega 1}^2 + E_{\omega 2}^2 - 2E_1E_{02}}$

$\sqrt{(E_{01} - E_{02})^2}$

$E_{Res} = \left[\,|E_{01} - E_{02}|\,,\ E_{01} + E_{02}\,\right]$ $[-1, 1]$

$I_{Res} = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos((\omega_2 - \omega_1)t + (\theta_2 - \theta_1))$

$I_{Res} = \left[\, I_1 + I_2 - 2\sqrt{I_1 I_2},\ I_1 + I_2 + 2\sqrt{I_1 I_2} \,\right]$

Eyes perceive

$I_{Avg} = I_1 + I_2$

Interference of incoherent waves

Ex

T_1.

$I_{tot} = 0.3 + 0.2 + 0.1$
$= 0.6 W/m^2$

Sun

T_2.

$I_1 = 0.3 W/m^2$ ✓

$I_2 = 0.2 W/m^2$ ✓

$I_3 = 0.1 W/m^2$ ✓

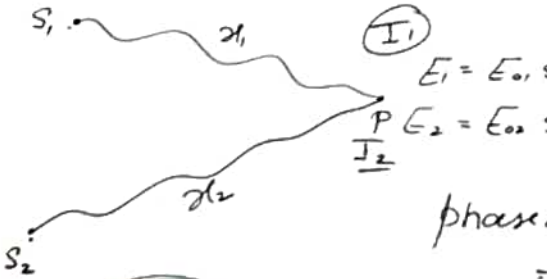
Interference of coherent sources

sources having same ω

$$E_{Res} = \sqrt{E_{01}^2 + E_{02}^2 + 2E_{01}E_{02}\cos\Delta\phi}$$

$$I_{Res} = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\Delta\phi$$

$$\neq f(t)$$



$E_1 = E_{01}\sin(\omega t - kx_1 + \theta_1)$

$E_2 = E_{02}\sin(\omega t - kx_2 + \theta_2)$

phase different

$$= \Delta\phi = k(x_2 - x_1) + (\theta_1 - \theta_2)$$

$$= k\Delta x + (\theta_1 - \theta_2)$$

$$= \frac{2\pi}{\lambda}\Delta x + (\theta_1 - \theta_2)$$

$k = \frac{\omega}{v}$ Δx = path difference

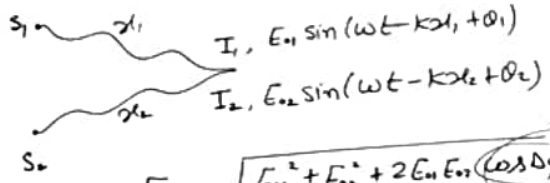
phase diff due to initial phases

in general $\theta_1 = \theta_2$

phase difference due to path difference

Interference of coherent sources

constructive interference



$$E_{Res} = \sqrt{E_{01}^2 + E_{02}^2 + 2E_{01}E_{02}\cos\Delta\phi}$$

E_{Res} → max

$\cos\Delta\phi \to +1$

$$\Delta\phi = 0, 2\pi, 4\pi, 6\pi, 2n\pi$$

$$(E_{Res})_{max} = \sqrt{E_{01}^2 + E_{02}^2 + 2E_1E_2 \times 1}$$

$$= (E_{01} + E_{02})$$

$$(I_{Res})_{max} = I_1 + I_2 + 2\sqrt{I_1 I_2}\cdot 1$$

$$= I_1 + I_2 + 2\sqrt{I_1 I_2}$$

$$= \left(\sqrt{I_1} + \sqrt{I_2}\right)^2$$

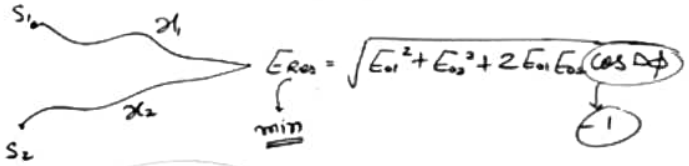
if $\theta_1 = \theta_2 \Rightarrow$

$$\frac{2\pi}{\lambda}(\Delta x) = 2n\pi$$

$$\Delta x = n\lambda = 0, \lambda, 2\lambda, 3\lambda, 4\lambda, 5\lambda \ldots$$

Interference of coherent sources

Destructive interference



$E_{Res} = \sqrt{E_{01}^2 + E_{02}^2 + 2E_{01}E_{02}\cos\Delta\phi}$ (min)

$E_{min} = \sqrt{E_{01}^2 + E_{02}^2 - 2E_{01}E_{02}}$

$= \sqrt{(E_{01} - E_{02})^2}$

$= |E_{01} - E_{02}|$

$\cos\Delta\phi = -1$

$\Delta\phi = \pi, 3\pi, 5\pi, 7\pi, \ldots (2n+1)\pi$

if $\theta_1 = \theta_2$

$\frac{2\pi}{\lambda}\Delta x = (2n+1)\pi$

$\Delta x = (2n+1)\frac{\lambda}{2}$ = odd $\times \frac{\lambda}{2} = \frac{\lambda}{2}, \frac{3\lambda}{2}, \frac{5\lambda}{2}, \frac{7\lambda}{2}, \frac{9\lambda}{2} - \cdots$

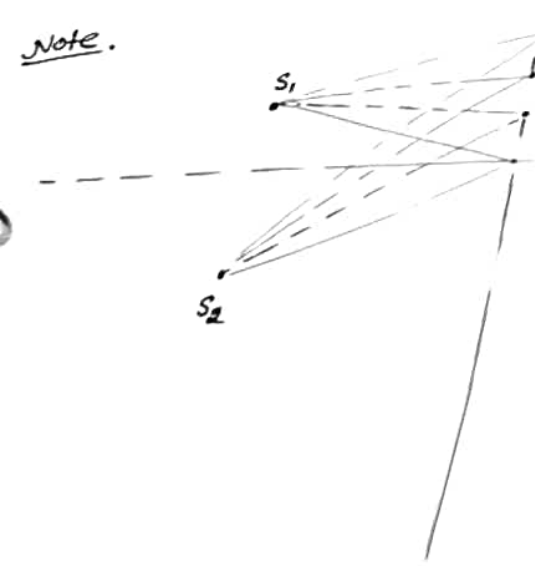
$I_{Res} = I_1 + I_2 + 2\sqrt{I_1 I_2} \times (-1)$

$= I_1 + I_2 - 2\sqrt{I_1 I_2}$

$I_{min} = (\sqrt{I_1} - \sqrt{I_2})^2$

Interference of coherent sources

Note.



$\Delta x = 1.5\lambda$ $\quad I_{res} = \left(\sqrt{I_1} - \sqrt{I_2}\right)^2$

$\Delta x = \lambda$ $\quad I_{res} = \left(\sqrt{I_1} + \sqrt{I_2}\right)^2$

$\Delta x = \frac{\lambda}{2}$ $\quad I_{res} = \left(\sqrt{I_1} - \sqrt{I_2}\right)^2$

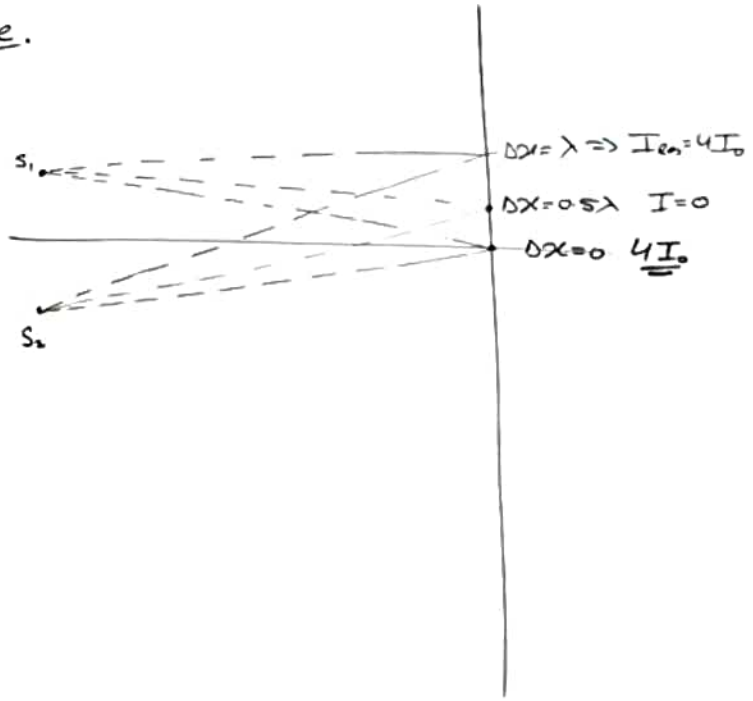
$\Delta x = 0$ $\quad I_{res} = \left(\sqrt{I_1} + \sqrt{I_2}\right)^2$

$$\text{Constrast} = \frac{I_{max}}{I_{min}} = \frac{\left(\sqrt{I_1} + \sqrt{I_2}\right)^2}{\left(\sqrt{I_1} - \sqrt{I_2}\right)^2}$$

Constrast "good" $I_1 = I_2$

Interference of coherent sources

Note.



$I_{Res} = I_0 + I_0 + 2I_0 \cos\Delta\phi$

$= 2I_0\{1 + \cos\Delta\phi\}$

$= 2I_0 \times 2\cos^2\frac{\Delta\phi}{2}$ $\quad \Delta\phi = \frac{2\pi}{\lambda}\Delta x$

$I_{Res} = 4I_0\cos^2\frac{\Delta\phi}{2}$ — [a1]

$I_{Res} \cdot [0, 4I_0]$

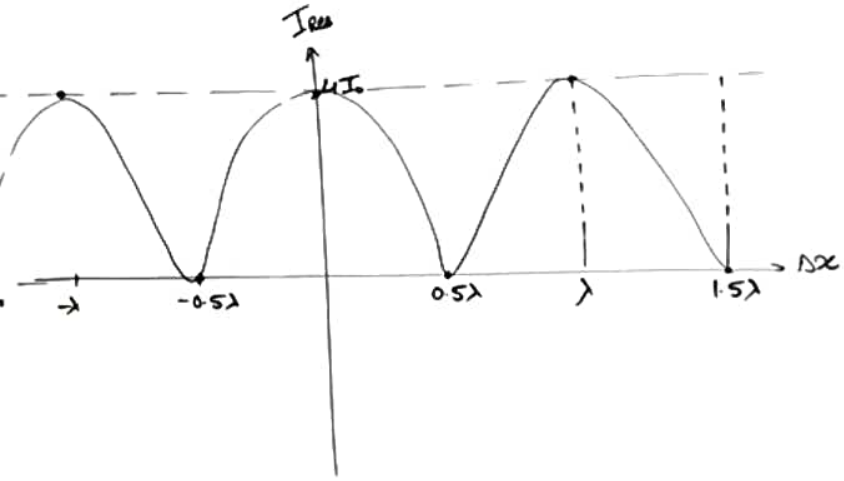
$I_{Res} = 4I_0\cos^2\frac{\pi}{\lambda}\Delta x$

$\Delta x = 0, \lambda, 2\lambda, 3\lambda, 4\lambda \ldots \Rightarrow I_{Res} = 4I_0$

$\Delta x = 0.5\lambda, 1.5\lambda, 2.5\lambda, \quad I_{Res} = 0$

Interference of coherent sources

Note.



$I_{res} = I_0 + I_0 + 2I_0 \cos\Delta\phi$

$= 2I_0\{1 + \cos\Delta\phi\}$

$= 2I_0 \times 2\cos^2\frac{\Delta\phi}{2}$ $\quad \Delta\phi = \frac{2\pi}{\lambda}\Delta x$

$I_{res} = 4I_0 \cos^2\frac{\Delta\phi}{2}$ — [a]

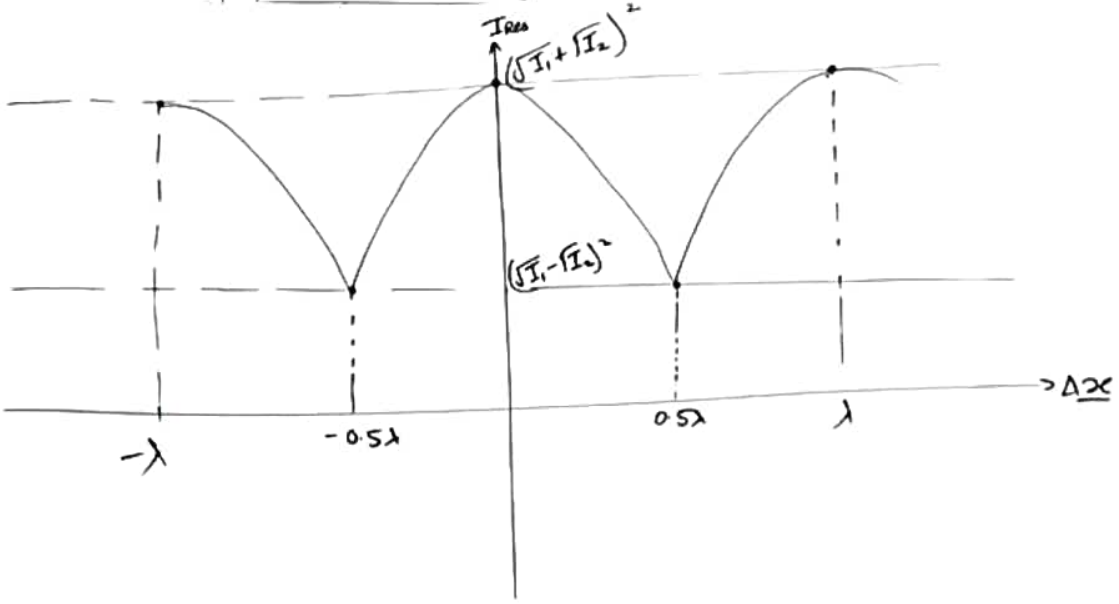
$I_{res} \cdot [0, 4I_0]$

$I_{res} = 4I_0 \cos^2\frac{\pi}{\lambda}\Delta x$

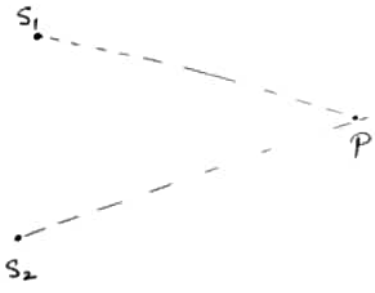
$\Delta x = 0, \lambda, 2\lambda, 3\lambda, 4\lambda \ldots \Rightarrow I_{res} = 4I_0$

$\Delta x = 0.5\lambda, 1.5\lambda, 2.5\lambda, \quad I_{res} = 0$

Interference of coherent sources



Ex



phase difference $= \Delta\phi = \frac{\pi}{3}$

find I_{Res} at point P.

$$I_{Res} = 0.3 + 0.4 + 2\sqrt{(0.3)(0.4)}\cos\frac{\pi}{3}$$

$$= 0.7 + \sqrt{0.12}$$

$$= 0.7 + \frac{\sqrt{12}}{10} = \left(0.7 + \frac{\sqrt{3}}{5}\right)\frac{Watt}{m^2}$$

Ex

S_1 $I_1 = 0.25 \frac{N}{m^2}$

S_2 $I_2 = 0.49 \frac{N}{m^2}$

find contrast $= \frac{1.44}{0.4} = 36$

find $I_{max} = \left(\sqrt{0.25} + \sqrt{0.49}\right)^2 = (0.5 + 0.7)^2 = 1.44\ N/m^2$

$I_{min} = \left(\sqrt{0.25} - \sqrt{0.49}\right)^2 = (0.5 - 0.7)^2 = 0.04\ N/m^2$

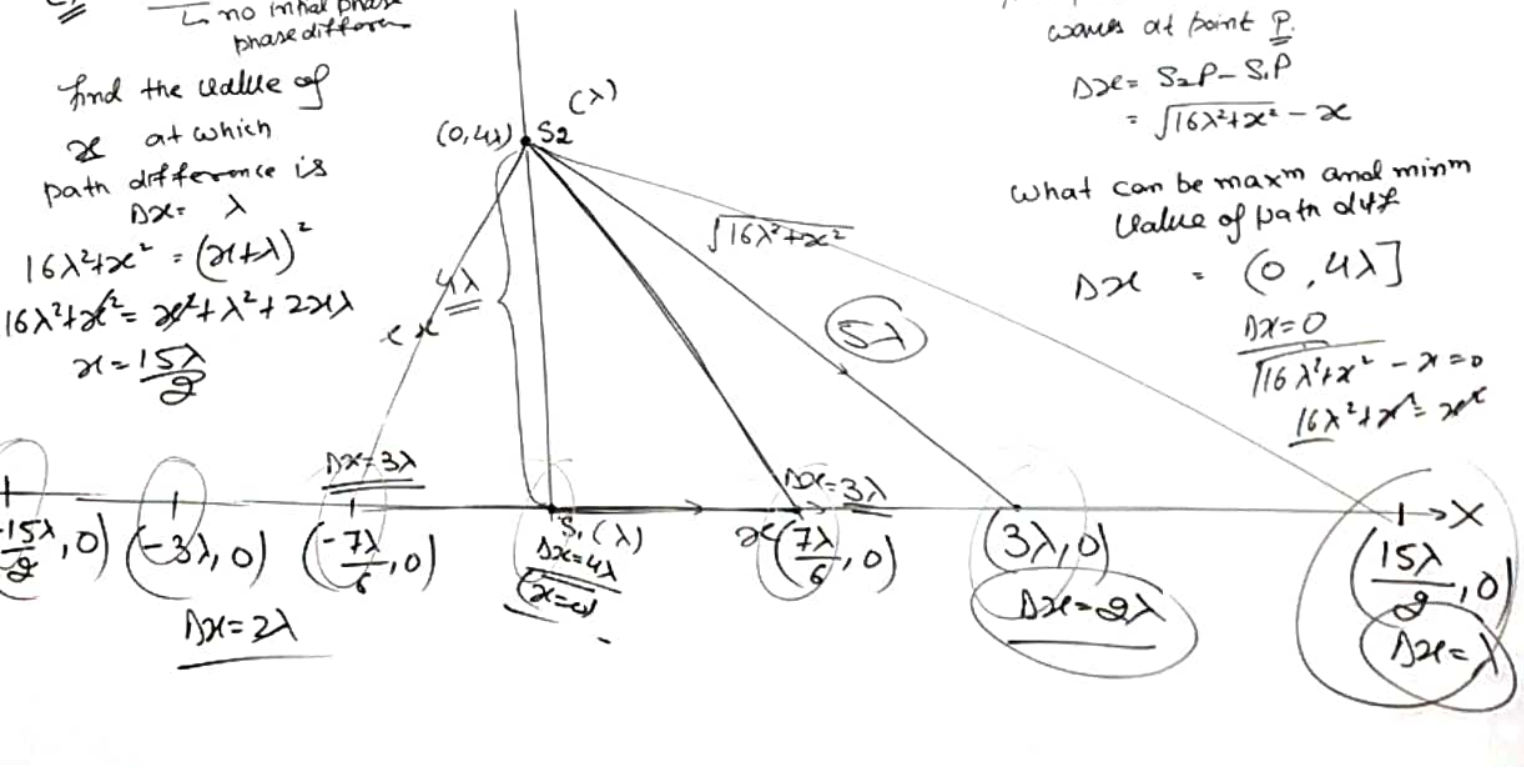
Ex S_1, S_2 coherent sources
↳ no initial phase
phase difference

find the value of x at which path difference is $\Delta x = \lambda$

$16\lambda^2 + x^2 = (x+\lambda)^2$

$16\lambda^2 + x^2 = x^2 + \lambda^2 + 2x\lambda$

$x = \frac{15\lambda}{2}$



find path difference b/w the waves at point P.

$\Delta x = S_2P - S_1P$

$= \sqrt{16\lambda^2 + x^2} - x$

what can be maxm and minm value of path diff

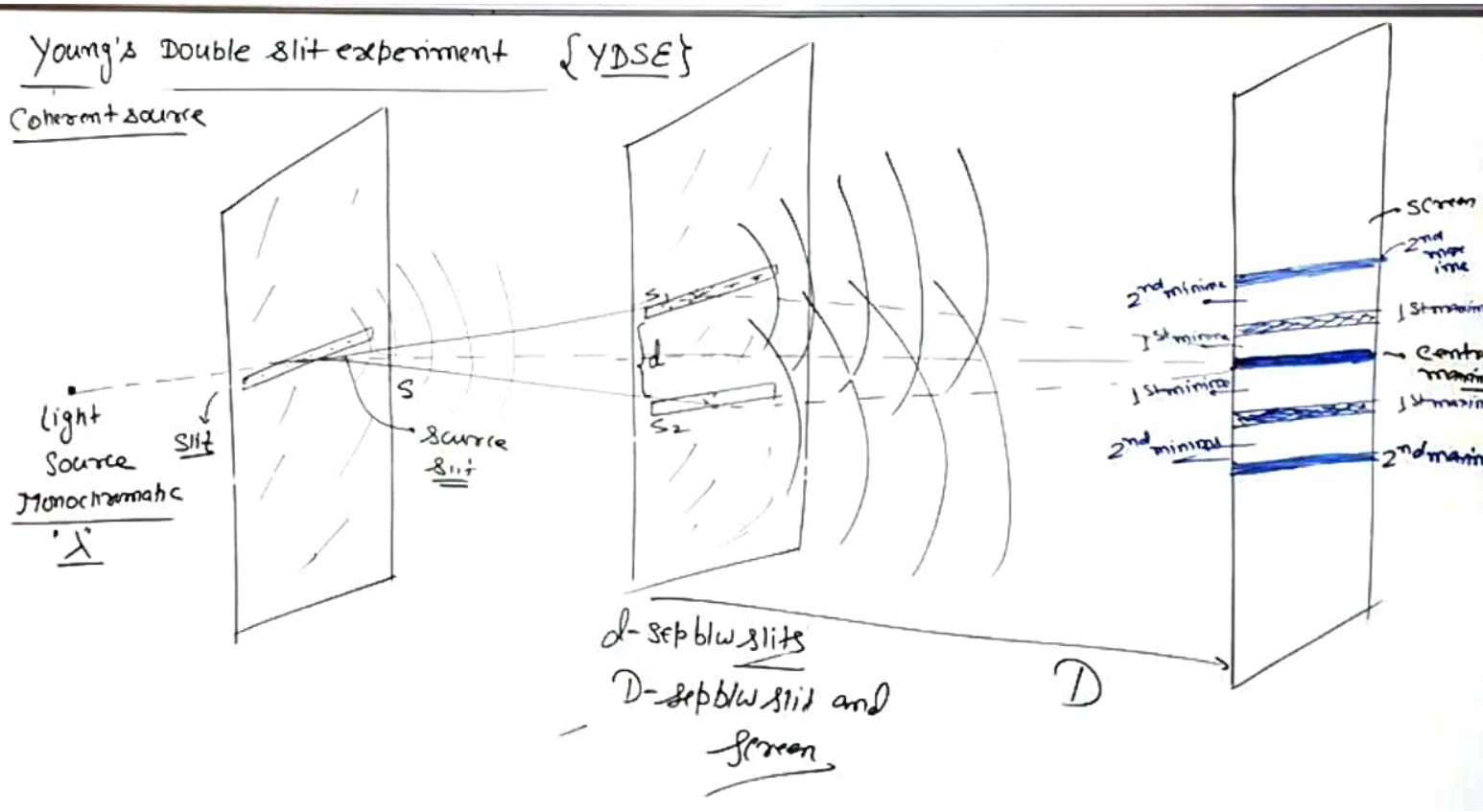
$\Delta x = (0, 4\lambda]$

$\Delta x = 0$

$\sqrt{16\lambda^2 + x^2} - x = 0$

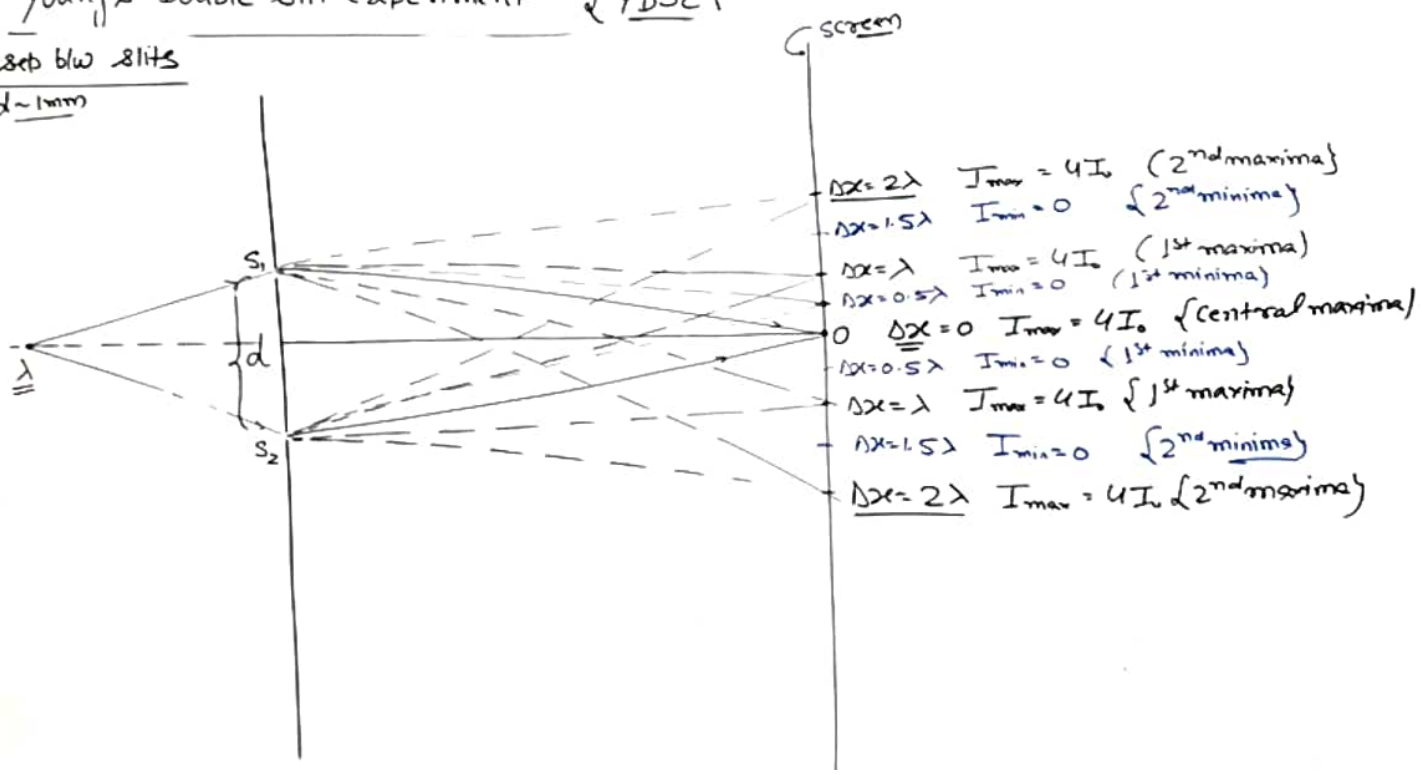
$16\lambda^2 + x^2 = x^2$

Young's Double slit experiment {YDSE}



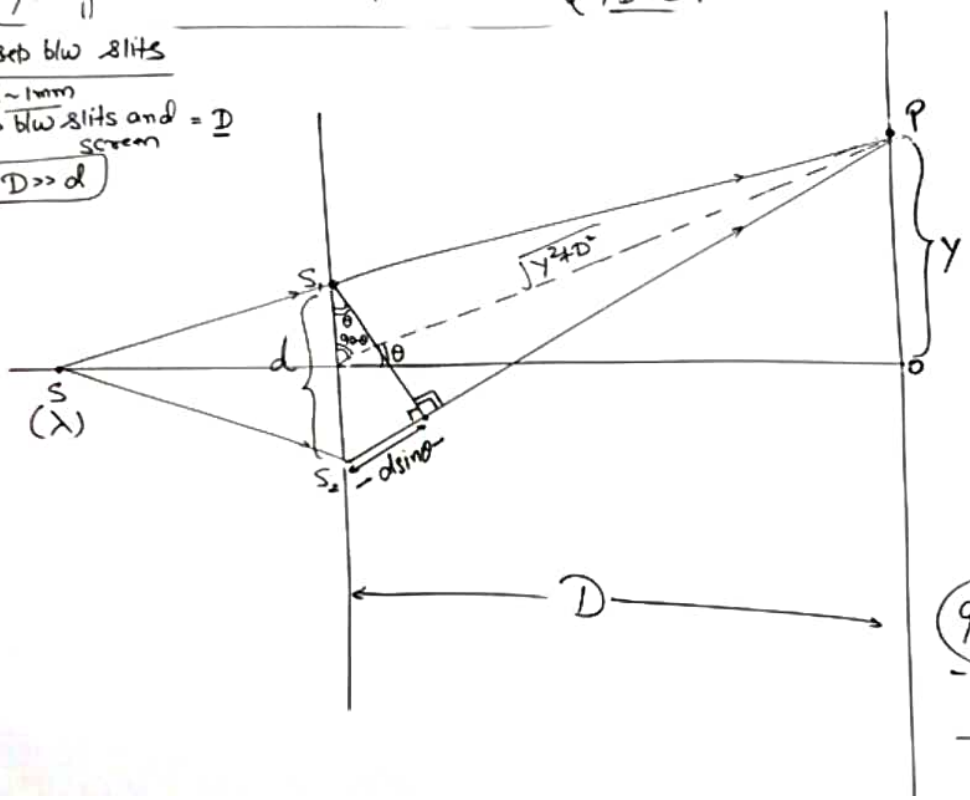
Young's Double slit experiment {YDSE}

d - sep b/w slits
d ~ 1mm



Young's Double slit experiment {YDSE}

d - Sep blw slits
d ~ 1mm
Sep blw slits and screen = D
D >> d



path difference
$= \Delta x = S_2P - S_1P$
$= d\sin\theta$ {Rem!}

$\Delta x = \frac{dY}{\sqrt{Y^2+D^2}}$

assumption
$Y << D$
$\theta \sim$ small
$\sin\theta \approx \tan\theta$

$\Delta x = d\sin\theta = d\tan\theta$

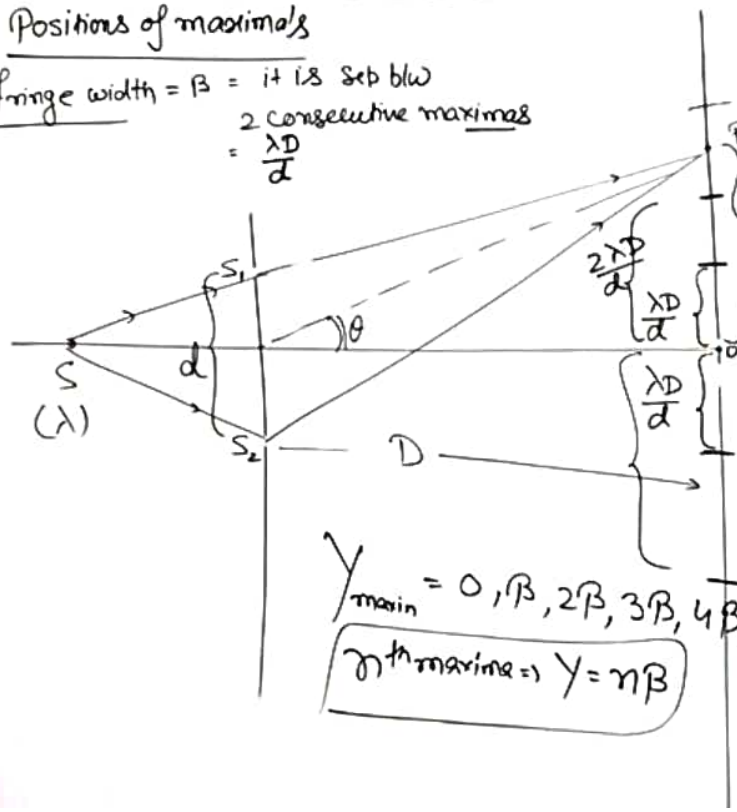
90%.
10%.

$\boxed{\Delta x = \frac{d.Y}{D}}$ Rem!

Young's Double slit experiment {YDSE}

Positions of maxima's

fringe width = β = it is seb b/w 2 consecutive maximas = $\frac{\lambda D}{d}$



Bright fringe (maxima)

$\Delta x = d\sin\theta \approx d\tan\theta = \frac{dy}{D} = n\lambda$

$y = \frac{n\lambda D}{d}$

$y = 0, \frac{\lambda D}{d}, \frac{2\lambda D}{d}, \frac{3\lambda D}{d}, \ldots$

n^{th} maxima.

$y = \frac{n\lambda D}{d}$

$y_{main} = 0, \beta, 2\beta, 3\beta, 4\beta, 5\beta$

n^{th} maxima $\Rightarrow y = n\beta$

$\frac{y}{D} = \frac{n\lambda}{d} << 1$

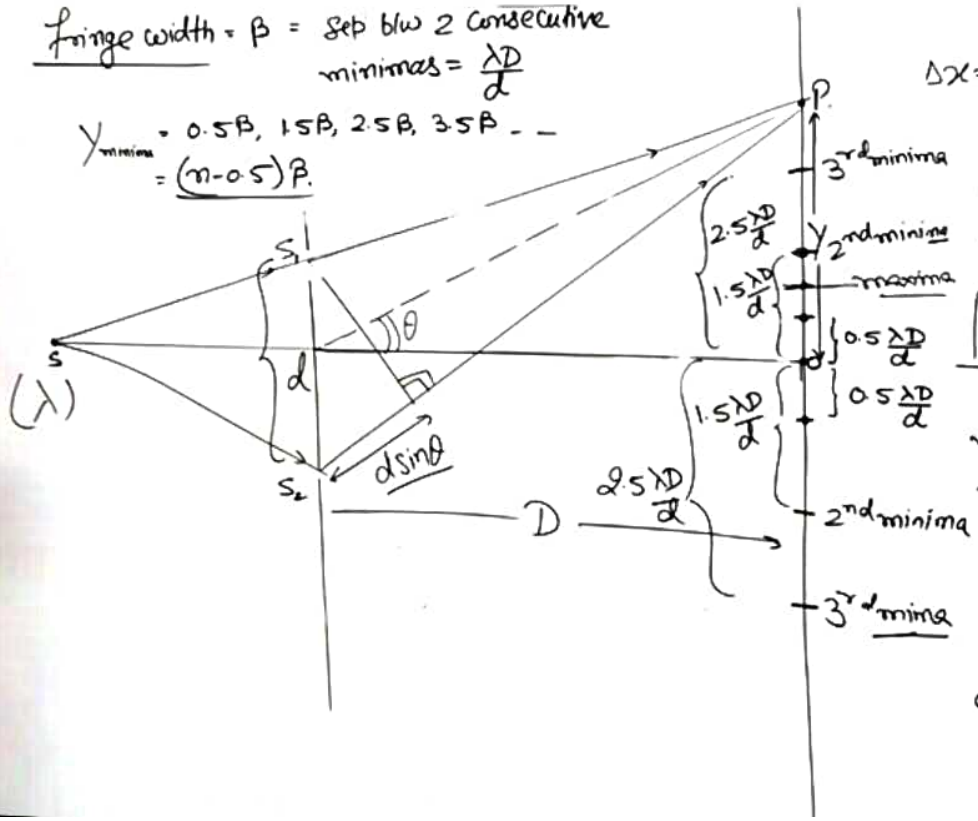
$n\lambda << d$

Young's Double slit experiment {YDSE}

Fringe width = β = Sep b/w 2 Consecutive minimas = $\frac{\lambda D}{d}$

y_{minima} = 0.5β, 1.5β, 2.5β, 3.5β - -

= $(n-0.5)\beta$.



$\Delta x = d\sin\theta \approx d\tan\theta = \frac{dy}{D} = (2n-1)\frac{\lambda}{2}$

$y = (2n-1)\frac{\lambda D}{2d}$

$y_{minima} = \left(n-\frac{1}{2}\right)\frac{\lambda D}{d}$

$y_{minima} = 0.5\frac{\lambda D}{d}, 1.5\frac{\lambda D}{d}, 2.5\frac{\lambda D}{d}, 3.5\frac{\lambda D}{d}, 4.5\frac{\lambda D}{d}$ - -

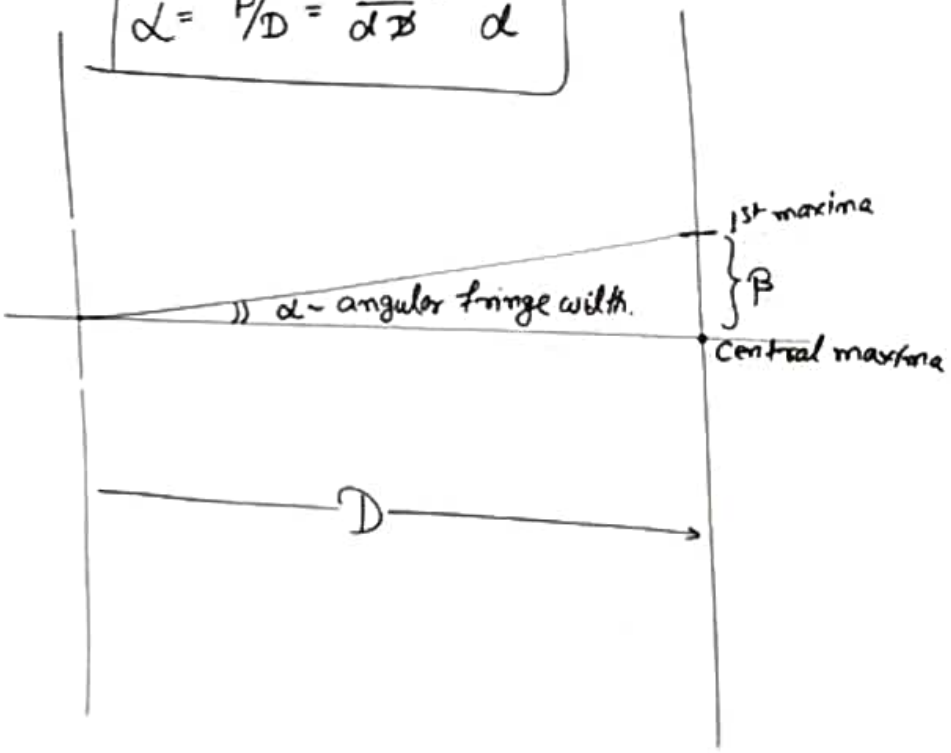
$(n-0.5)\frac{\lambda D}{d}$

8th minima. $y = 7.5\frac{\lambda D}{d}$

Young's Double slit experiment {YDSE}

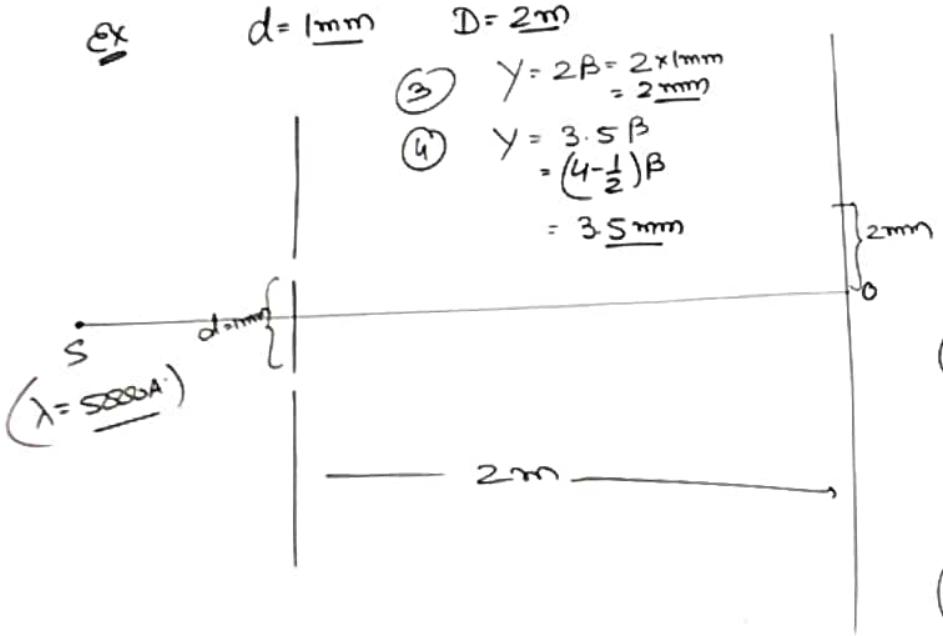
Angular fringe width

$$\alpha = \beta/D = \frac{\lambda D}{d D} = \frac{\lambda}{d}$$



Young's Double slit experiment {YDSE}

Ex $d = 1mm$ $D = 2m$



③ $Y = 2\beta = 2 \times 1mm = 2mm$

④ $Y = 3.5\beta = (4 - \frac{1}{2})\beta = 3.5mm$

Find ① Fringe width

② Angular fringewidth

③ position of 2nd maxima

④ position of 4th minima

① $\beta = \frac{\lambda D}{d} = \frac{5000 \times 10^{-10} \times 2}{10^{-3}} m$

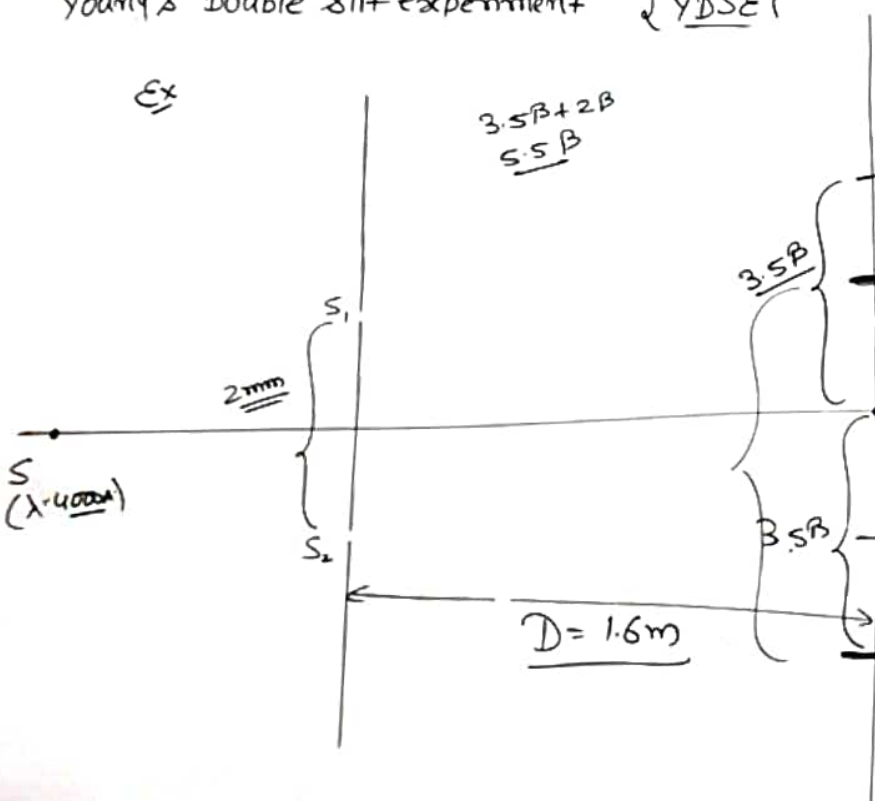
$= 10^{-3} m = 1mm$

② $\alpha = \frac{\lambda}{d} = \frac{\beta}{D} = \frac{10^{-3} m}{2m}$

$= 5 \times 10^{-4}$ Rad

Young's Double slit experiment {YDSE}

Ex



Find Separation btw 2nd maxima and 4th minima

$\beta = \frac{\lambda D}{d}$

$\left(4 - \frac{1}{2}\right)\beta = 3.5\beta$

$1.5\beta = 1.5\,\frac{\lambda D}{d}$

$= 1.5 \times \frac{4000 \times 10^{-10} \times 1.6}{2 \times 10^{-3}}\ m$

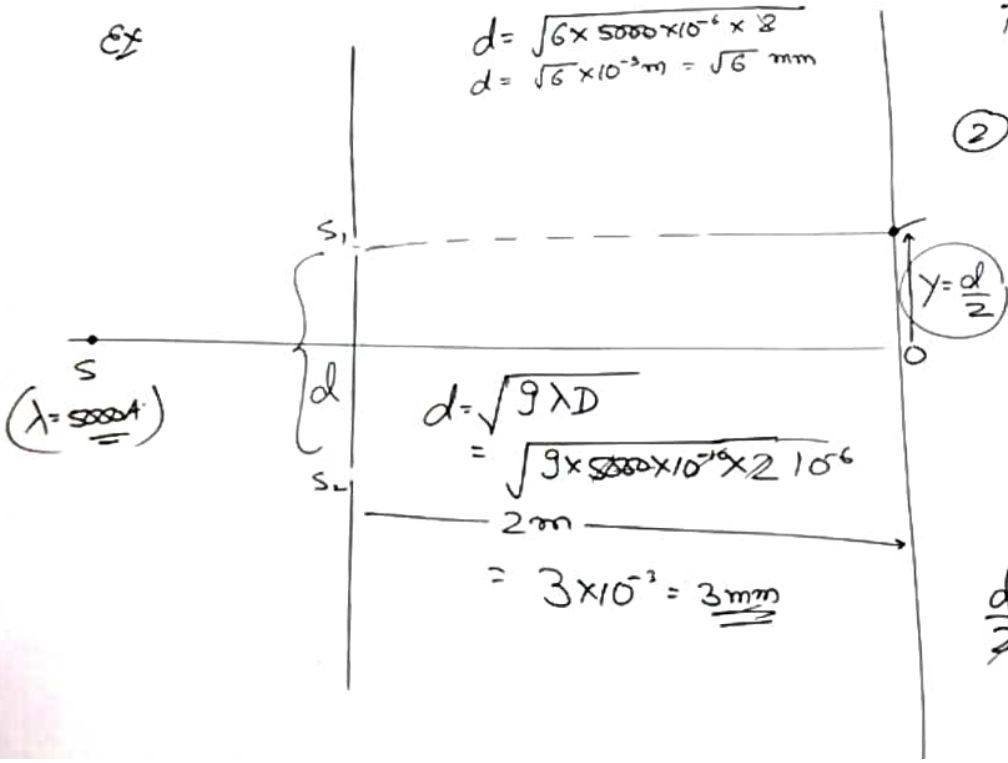
$= 4.8 \times 10^{-4}\ m$

Young's Double slit experiment {YDSE}

Ex

$d = \sqrt{6 \times 5000 \times 10^{-6} \times 8}$

$d = \sqrt{6} \times 10^{-3} m = \sqrt{6}$ mm



$d = \sqrt{9 \lambda D}$

$= \sqrt{9 \times 5000 \times 10^{-10} \times 2 \times 10^{6}}$

$= 3 \times 10^{-3} = 3mm$

Find separation b/w slits such that

(2) in front of S_1 on the screen, there is 5th minima.

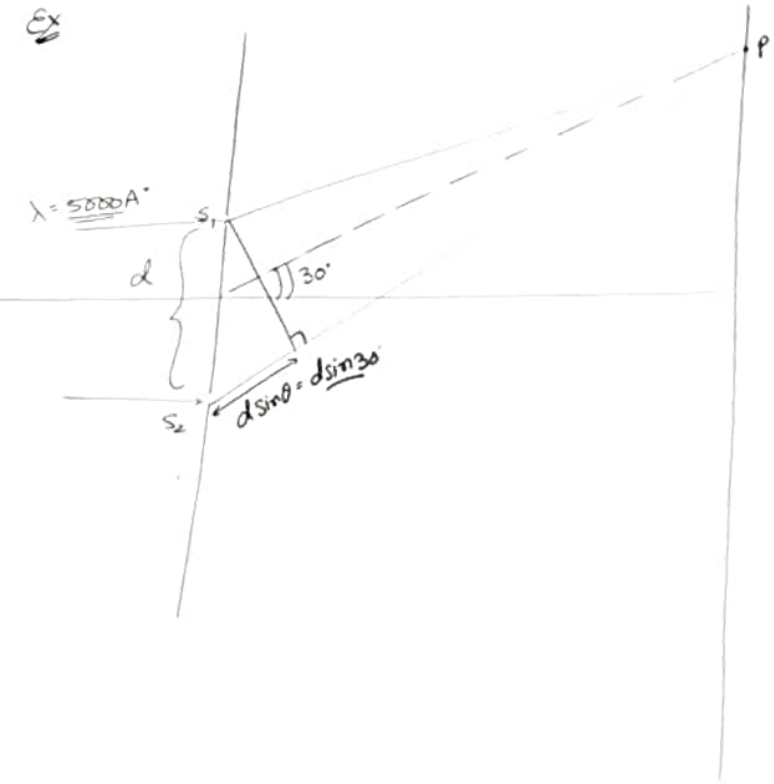
$\Delta x = \frac{dy}{D}$

$= \frac{d \cdot d}{2D} = \frac{d^2}{2D} \cdot \frac{9}{2} \lambda$

$d = \sqrt{9 \lambda D}$

$\frac{d}{2} = y = 4.5 \beta = \frac{9}{2} \frac{\lambda D}{d}$

Ex



Find 'd' so that at point P there is 5th maxima.

$\Delta x = d\sin\theta = \left(n-\frac{1}{2}\right)\lambda$

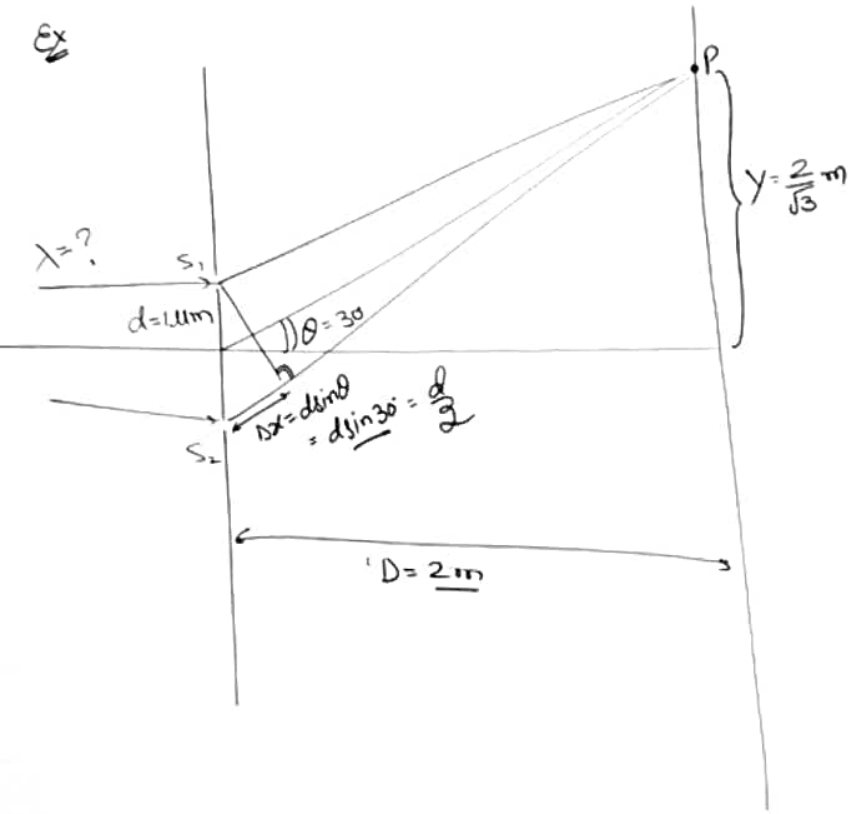
$d\sin 30^\circ = 4.5\lambda$

$d = 9\lambda$

$= 9 \times 5000A^\circ$

$= 45,000A^\circ$

Ex



find 'λ' of EM so that there is 2nd minima at point P.

$$\tan\theta = \frac{2/\sqrt{3}}{2} = \frac{1}{\sqrt{3}}$$

$$\theta = 30^\circ$$

$$\Delta x = \frac{d}{2} = 1.5\lambda = \frac{3\lambda}{2}$$

$$\lambda = \frac{d}{3} = \frac{10^{-6}\,m}{3}$$

$$= 3.333 \times 10^{-7}\,m$$

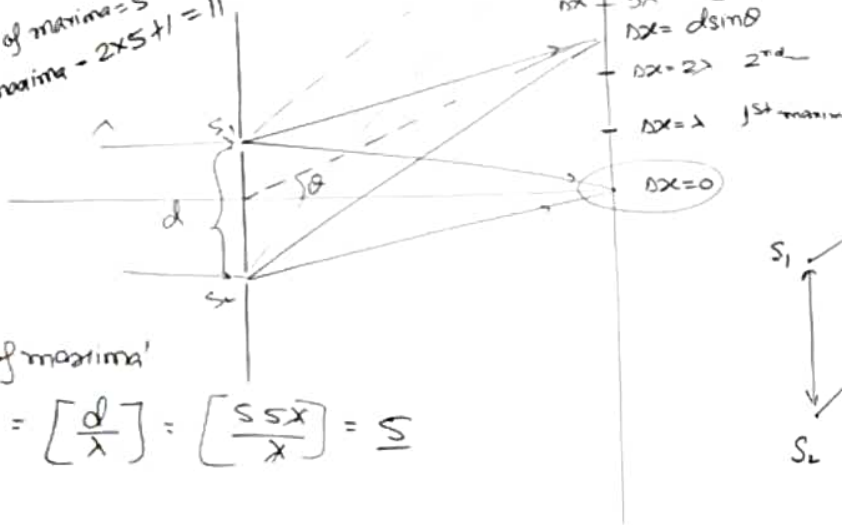
$$= 3333\,A^\circ$$

Highest order of maxima

$d = 5.5\lambda$

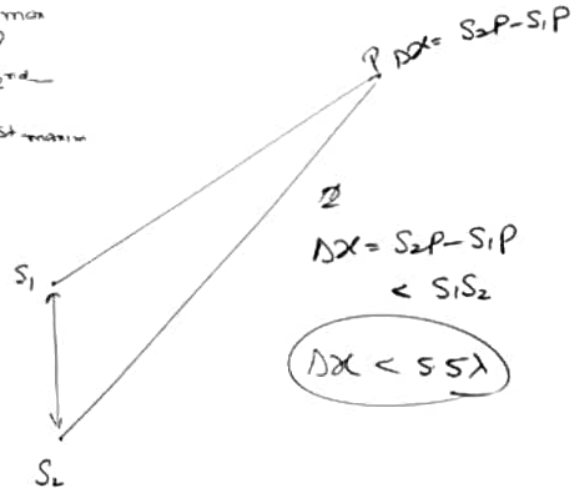
Highest order of maxima = 5

no of maxima = $2 \times 5 + 1 = 11$





$S_2P - S_1P = 5\lambda$



$\Delta x = S_2P - S_1P$

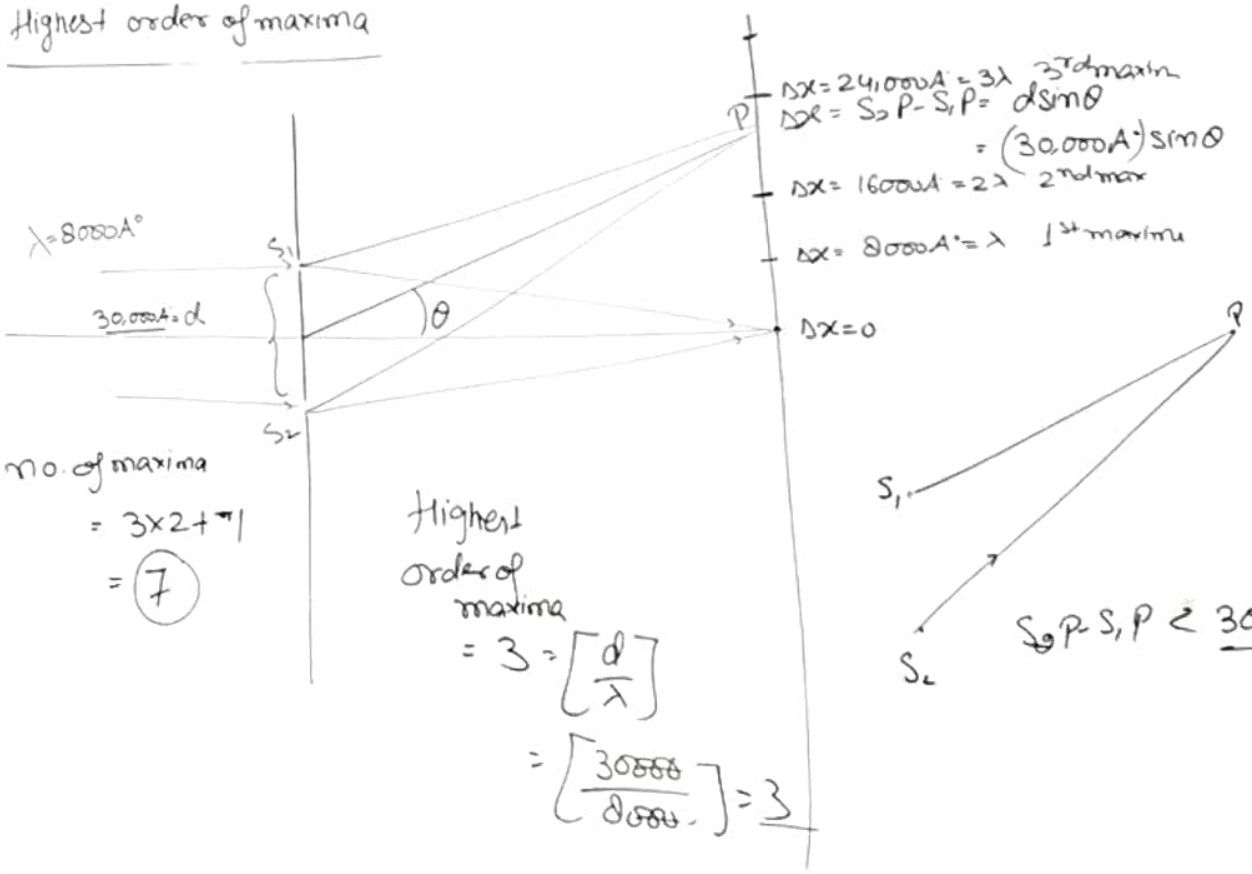
$< S_1S_2$

$\Delta x < 5.5\lambda$

Order of maxima

$= \left[\frac{d}{\lambda}\right] = \left[\frac{5.5\lambda}{\lambda}\right] = 5$

Highest order of maxima



no. of maxima

= 3x2+1

= (7)

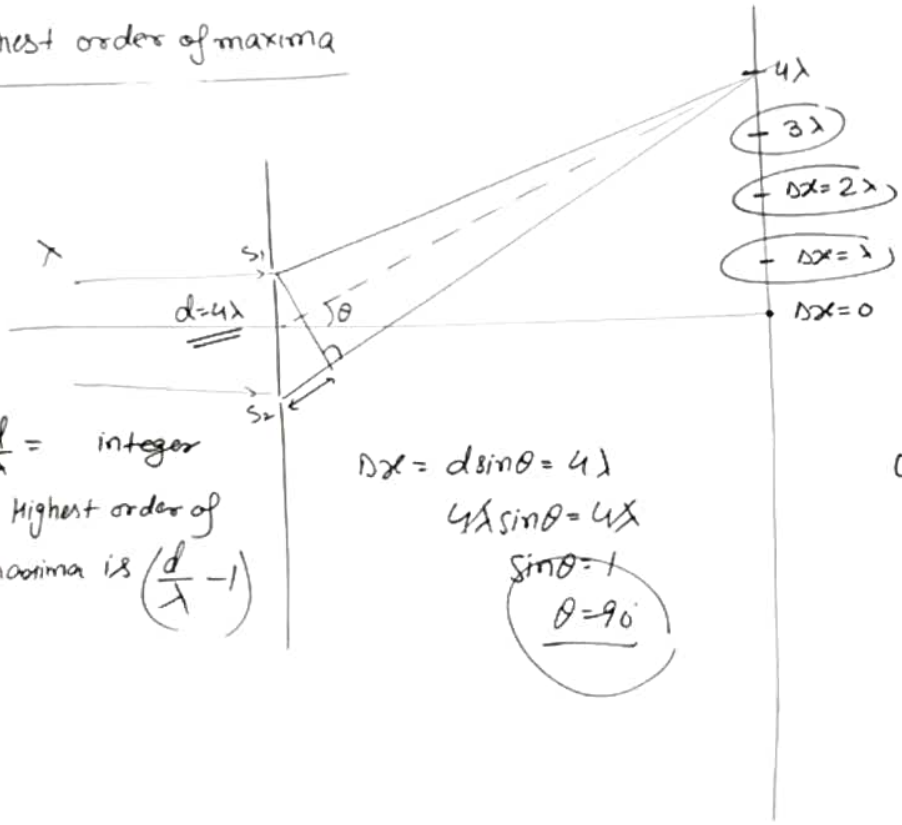
Highest order of maxima

$= 3 = \left[\frac{d}{\lambda}\right]$

$= \left[\frac{30000}{8000}\right] = 3$

$S_2P - S_1P < 30000A°$

Highest order of maxima



3 Maximum order

order of maxima = 3

no. of maxima's = 7

If $\frac{d}{\lambda}$ = integer

then Highest order of maxima is $\left(\frac{d}{\lambda} - 1\right)$

$\Delta x = d \sin\theta = 4\lambda$

$4\lambda \sin\theta = 4\lambda$

$\sin\theta = 1$

$\theta = 90^\circ$

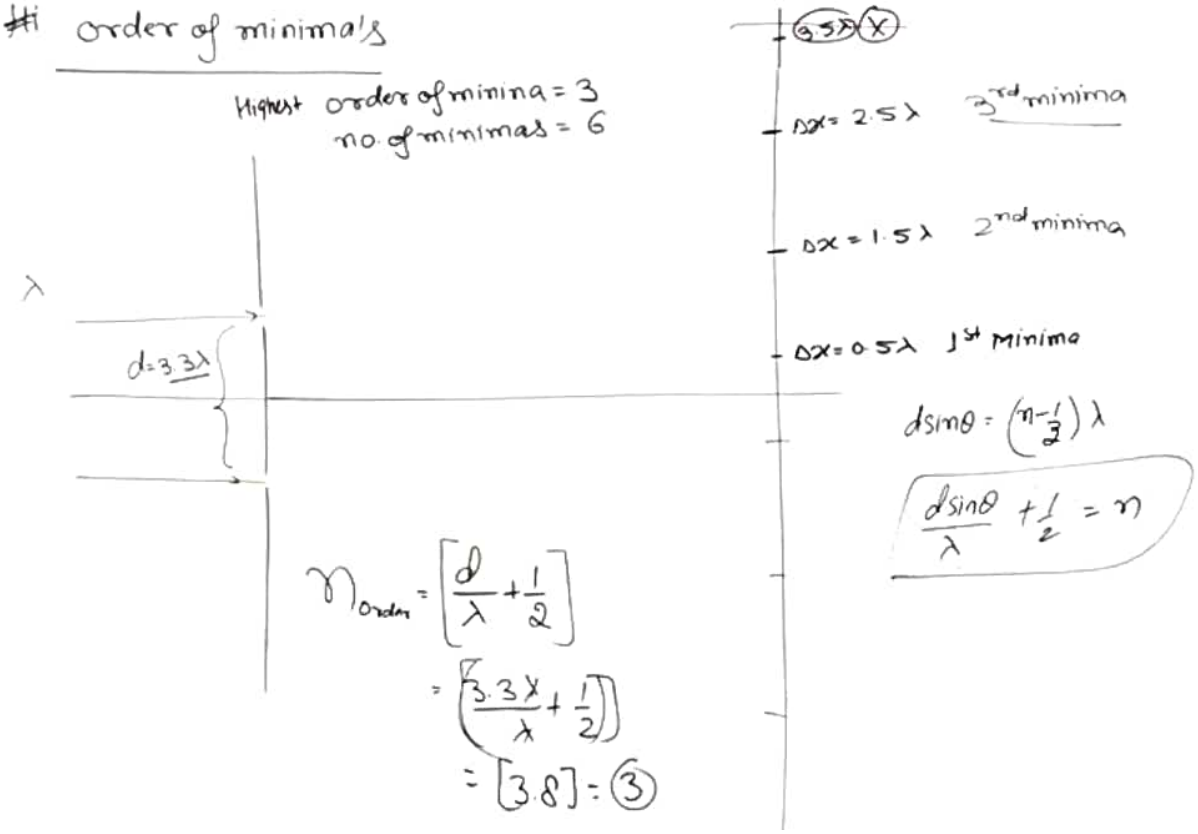
Order of maxima = $\left[\frac{d}{\lambda}\right]$

$= \left[\frac{4\lambda}{\lambda}\right]$

$= 4$

#i Order of minima's

Highest order of minima = 3
no. of minimas = 6



$$d\sin\theta = \left(n - \frac{1}{2}\right)\lambda$$

$$\frac{d\sin\theta}{\lambda} + \frac{1}{2} = n$$

$$n_{order} = \left[\frac{d}{\lambda} + \frac{1}{2}\right]$$

$$= \left[\left[\frac{3.3\lambda}{\lambda} + \frac{1}{2}\right]\right]$$

$$= [3.8] = 3$$

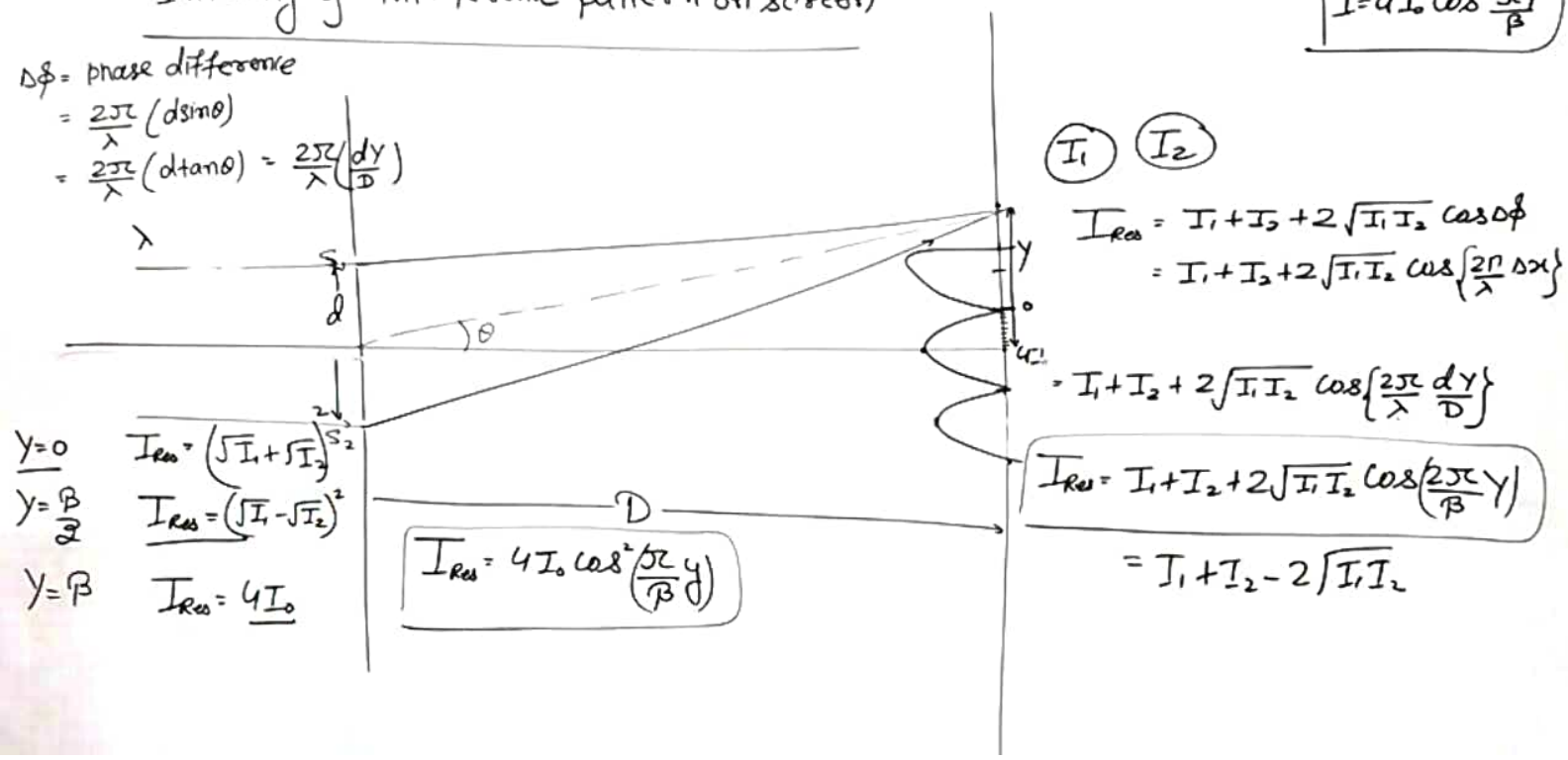
Intensity of interference pattern on screen

$I = 4I_0 \cos^2 \frac{\pi y}{\beta}$

$\Delta\phi$ = phase difference

$= \frac{2\pi}{\lambda}(d\sin\theta)$

$= \frac{2\pi}{\lambda}(d\tan\theta) = \frac{2\pi}{\lambda}\left(\frac{dy}{D}\right)$



$I_{Res} = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\Delta\phi$

$= I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\left\{\frac{2\pi}{\lambda}\Delta x\right\}$

$= I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\left\{\frac{2\pi}{\lambda}\frac{dy}{D}\right\}$

$I_{Res} = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\left(\frac{2\pi}{\beta}y\right)$

$= I_1 + I_2 - 2\sqrt{I_1 I_2}$

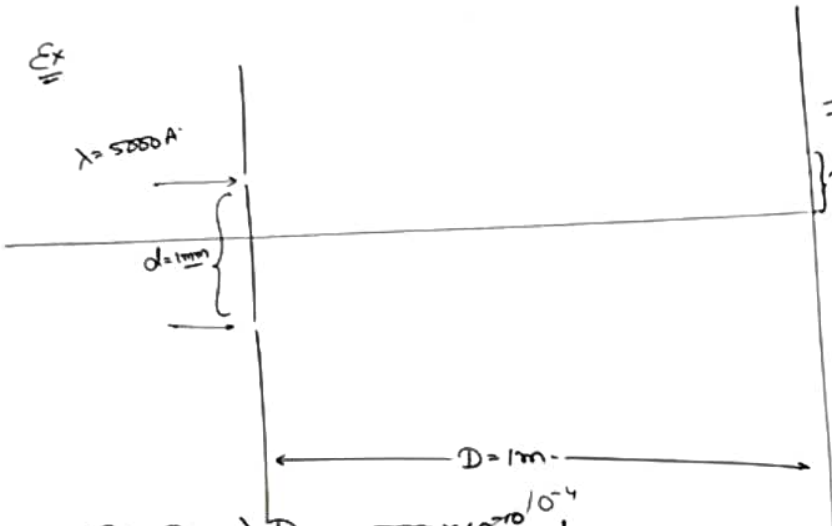
$y = 0$ $\quad I_{Res} = \left(\sqrt{I_1} + \sqrt{I_2}\right)^2$

$y = \frac{\beta}{2}$ $\quad I_{Res} = \left(\sqrt{I_1} - \sqrt{I_2}\right)^2$

$y = \beta$ $\quad I_{Res} = 4I_0$

$I_{Res} = 4I_0\cos^2\left(\frac{\pi}{\beta}y\right)$

Ex



Find

① Fringe width

② position of 3rd maxima

③ position of 5th minima

④ intensity at maxima ✓

⑤ -- ——— minima

④ $I_{max} = \left(\sqrt{I_0} + \sqrt{4I_0}\right)^2$

$= I_0(9) = 9I_0$ ✓

Contrast $= \frac{9I_0}{I_0} = 9$

① $\beta = \frac{\lambda D}{d} = \frac{5000 \times 10^{-10} \times 1}{10^{-3}}$ m

$= 0.5$ mm

② $y = 3\beta = 1.5$ mm

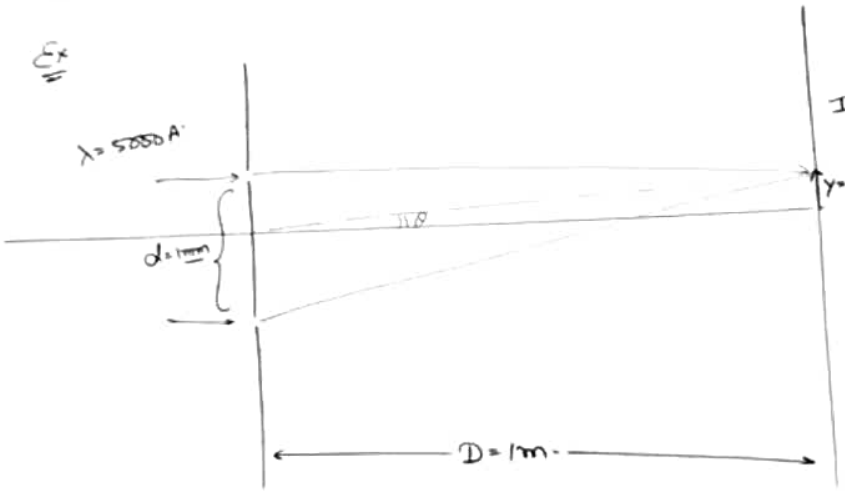
③ $y = 4.5\beta$

$= 4.5 \times 0.5 = 2.25$ mm

⑤ $I_{min} = \left(\sqrt{I_0} - \sqrt{4I_0}\right)^2$

$= I_0 \times 1 = I_0$ ✓

Ex



find

(6) path difference at $y = \frac{1}{3}$ mm

(7) phase difference at $y = \frac{1}{3}$ mm

(8) I_{Res} at $y = \frac{1}{3}$

$$I_{Res} = I_0 + 4I_0 + 2\sqrt{I_0 4I_0}\cos\frac{4\pi}{3}$$

$$= 5I_0 + 4I_0 \times \left\{-\cos\frac{\pi}{3}\right\}$$

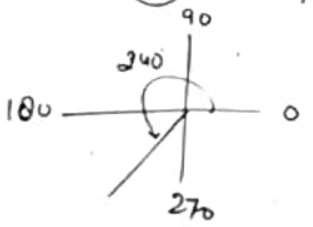
$$= 5I_0 + 4I_0\left(-\frac{1}{2}\right)$$

$$\boxed{I_{Res} = 3I_0}$$

(7) $\Delta\phi = \frac{2\pi}{\lambda}\Delta x$

$$= \frac{2\pi}{5000\times10^{-10}} \times \frac{10^{-6}}{3} \times 2 = \frac{4\pi}{3}$$

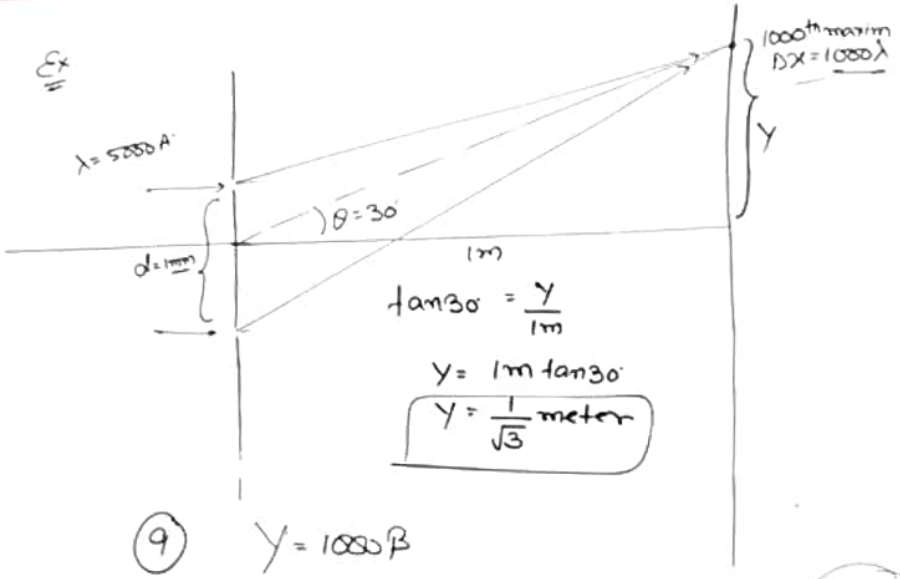
$$= 240^\circ$$



$\cos(\pi+\theta) = -\cos\theta$

$\cos\left(\frac{4\pi}{3}\right) = \cos\left(\pi+\frac{\pi}{3}\right) = -\cos\frac{\pi}{3}$

Ex



$\tan 30^\circ = \frac{Y}{1m}$

$Y = 1m \tan 30^\circ$

$Y = \frac{1}{\sqrt{3}}$ meter

find

(6) path difference at $y = \frac{1}{3}$ mm

(7) phase difference at $y = \frac{1}{3}$ mm

(8) I_{Res} at $y = \frac{1}{3}$

(9) position of 1000th maxima

(10) ---- 5000th maxima

(11) total no. of maximas

(9) $Y = 1000\beta$

$= 1000 \times 0.5$ mm

$= 0.5$m X

$y << D$ X

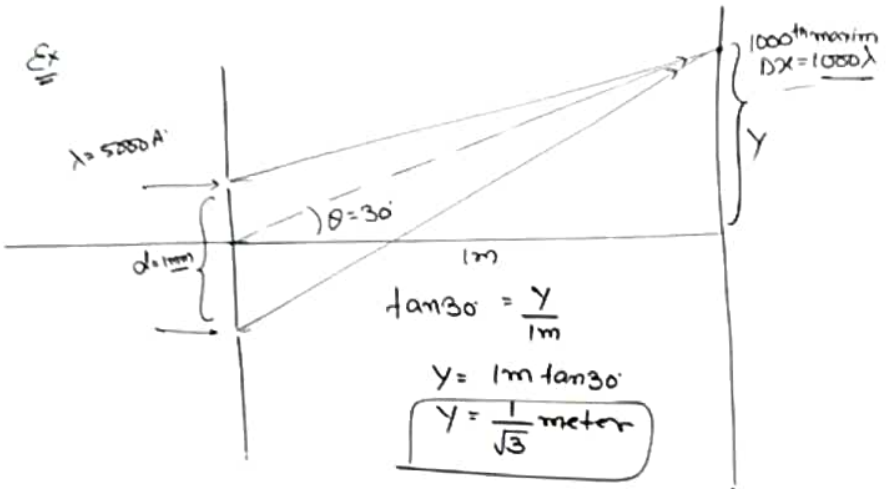
$\theta \sim$ small X

$\Delta x = d\sin\theta = 1000\lambda$

$\sin\theta = \frac{1000 \times 5000 \times 10^{-10}}{10^{-3}}$ $10^{-1} = \frac{5}{10} = \frac{1}{2}$

$\theta = 30^\circ$

Ex 3



$\tan 30^\circ = \frac{Y}{1m}$

$Y = 1m \tan 30^\circ$

$Y = \frac{1}{\sqrt{3}}$ meter

find

(6) path difference at $y = \frac{1}{3}$ mm

(7) phase difference at $y = \frac{1}{3}$ mm

(8) I_{res} at $y = \frac{1}{3}$

(9) position of 1000th maxima

(10) — — — 5000th maxima

(11) total no. of maximas

(11) Highest order of maximas = $\frac{d}{\lambda} - 1$

$\frac{d}{\lambda} = \frac{10^{-3} \times 10^{4}}{5000 \times 10^{-10} \times 10^{4}} = 2000$

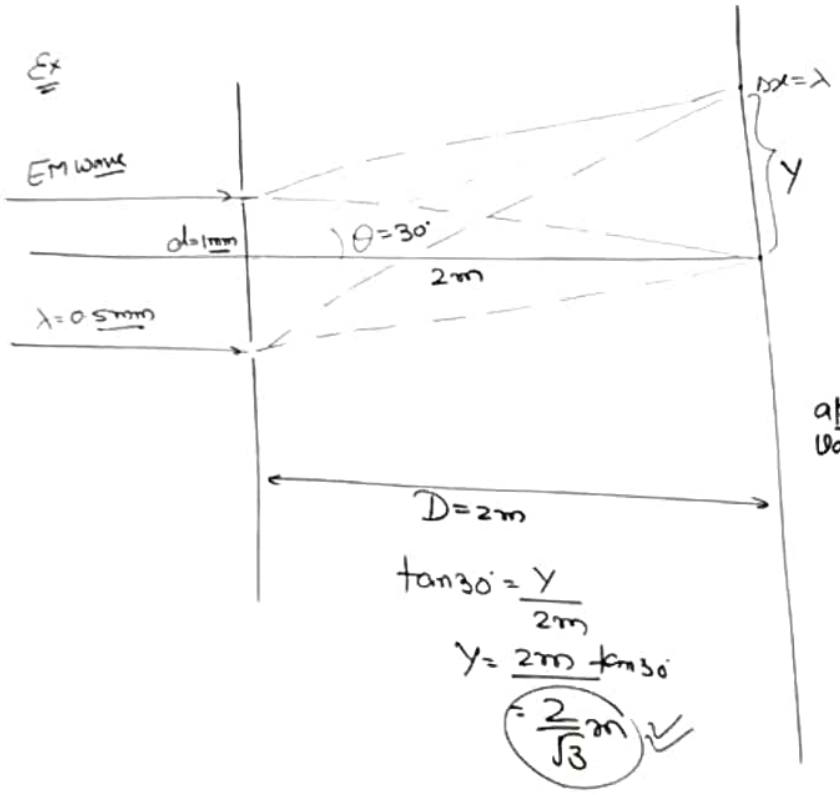
$d \sin\theta = 2000\lambda$

$\sin\theta = \frac{2000 \times 5000 \times 10^{-10} \times 10^{4}}{10^{3}} = \frac{10}{10} = 1$ $\theta = 90^\circ$

1999 (4001) ✓

$1 + 2 \times 1999 = 3999$ ✓

Ex



$\tan 30^\circ = \frac{y}{2m}$

$y = 2m \tan 30^\circ$

$= \frac{2}{\sqrt{3}} m$

find sep btw central maxima and 1st maxima.

$\beta = \frac{\lambda D}{d}$

$= \frac{0.5 \times 10^{-3} \times 2}{10^{-3}} = 1m \not< 2m$

Essential Multi Ex: I II AB

approx Validation $n\lambda << d$

$1 \times 0.5mm \not< 1mm$

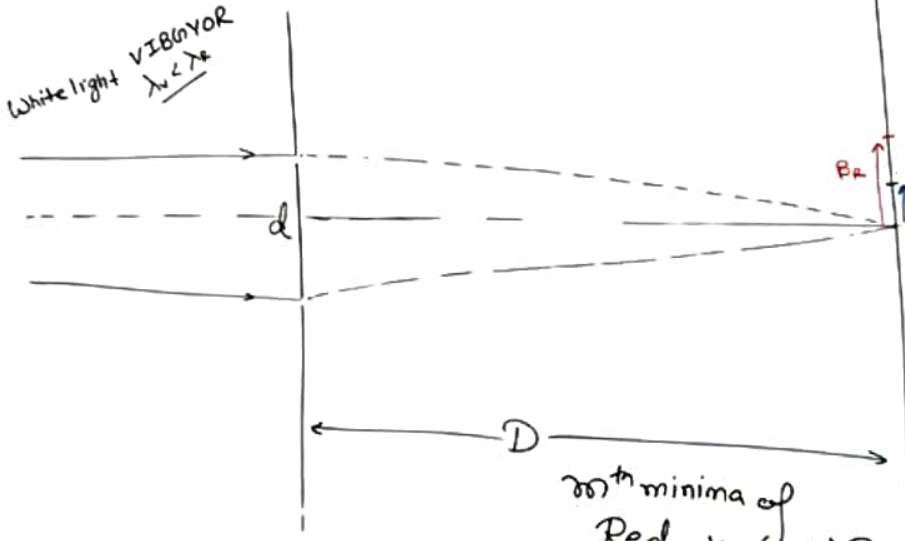
$\Delta x = d \sin\theta = n\lambda$

$\sin\theta = \frac{1 \times 0.5 \times 10^{-3}}{10^{-3}} = \frac{1}{2}$

$\theta = 30^\circ$

YDSE with Polychromatic light / white light.

↳ many λ's



n^{th} maxima of violet

$$y = n\beta_v = n\frac{\lambda_v D}{d}$$

n^{th} maxima of Red

$$y = n\beta_R = n\frac{\lambda_R D}{d}$$

$$\beta_v = \frac{\lambda_v D}{d}$$

$$\beta_R = \frac{\lambda_R D}{d}$$

n^{th} minima of Red $= y = \left(n - \frac{1}{2}\right)\beta_R$

$$= \left(n - \frac{1}{2}\right)\frac{\lambda_R D}{d}$$

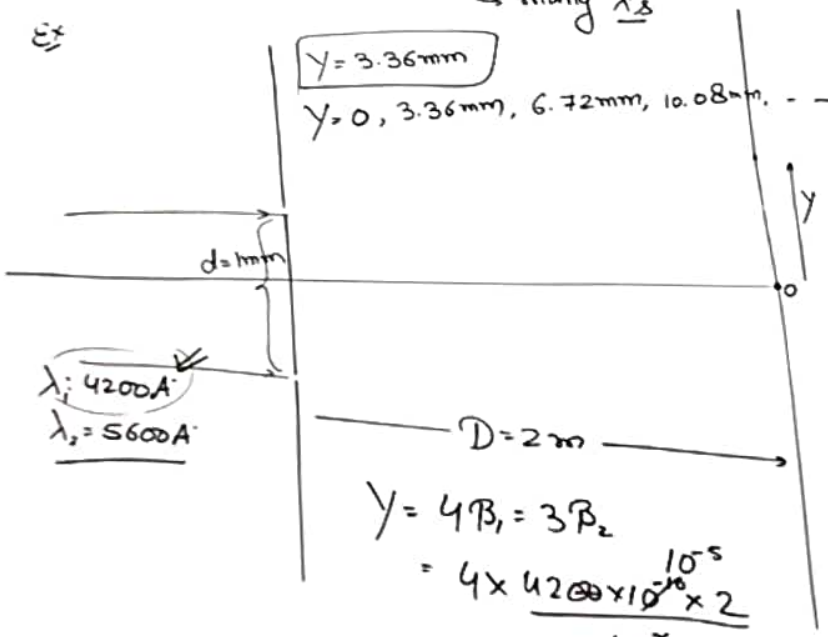
n^{th} minima of violet

$$y = \left(n - \frac{1}{2}\right)\beta_v = \left(n - \frac{1}{2}\right)\frac{\lambda_v D}{d}$$

YDSE with Polychromatic light / White light.

→ many λ's

Ex



Find the positions on screen where maxima's of these wavelengths will coincide.

$y = n_1\beta_1 = n_2\beta_2$

$\frac{n_1 \lambda_1 D}{d} = \frac{n_2 \lambda_2 D}{d}$

$\frac{n_1}{n_2} = \frac{\lambda_2}{\lambda_1} = \frac{56}{42} = \frac{4}{3}$

$n_1 = 4$, $n_2 = 3$ ✓

$n_2 = 8$, $n_1 = 6$

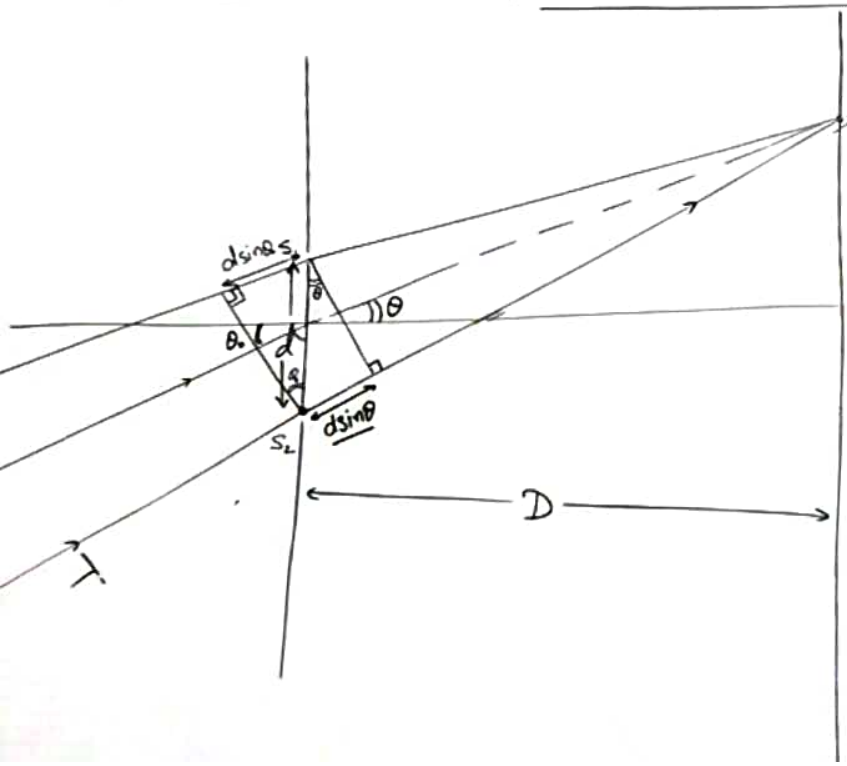
$n_2 = 12$, $n_1 = 9$

$y = 4\beta_1 = 3\beta_2$

$= 4 \times \frac{4200 \times 10^{-10} \times 2}{10^{-3}}$

$= 42 \times 8 \times 10^{-5} m = 336 \times 10^{-5} m = 3.36 mm$

Oblique incidence of light / Changing dirn of incident light



2nd maxima => $y = \frac{2\lambda D}{d} + D\theta_0$ $\quad y = 2\beta + D\theta_0$

$\Delta x = d\sin\theta - d\sin\theta_0$

$= \frac{dy}{D} - d\sin\theta_0$

maxima's

θ_0 ~ small

Radian

$\Delta x = \frac{dy}{D} - d\theta_0 = n\lambda$

$\frac{dy}{D} = n\lambda + d\theta_0$

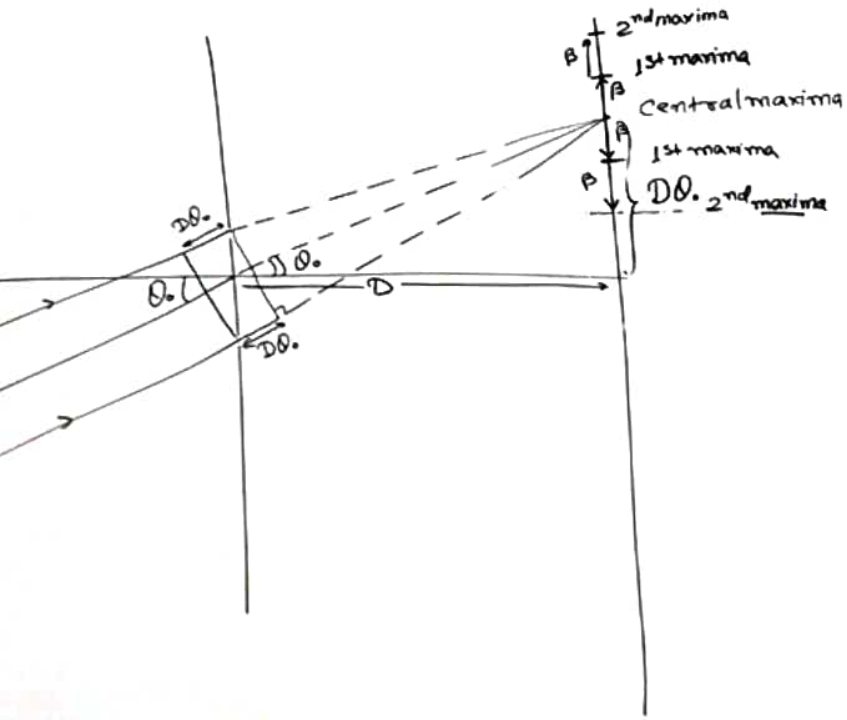
$y = \frac{n\lambda D}{d} + D\theta_0$

Central maxima $n = 0$ $\quad y = D\theta_0$

1st maxima => $n = 1$ $\quad y = \frac{\lambda D}{d} + D\theta_0$

$= \beta + D\theta_0$

Oblique incidence of light / changing dirn of incident light



2nd maxima $\Rightarrow$ $y = \frac{2\lambda D}{d} + D\theta_0$

$y = 2\beta + D\theta_0$

$\Delta x = d\sin\theta - d\sin\theta_0$

$= \frac{dy}{D} - d\sin\theta_0$

maxima's

θ_0 ~ small (Radian)

$\Delta x = \frac{dy}{D} - d\theta_0 = n\lambda$

$\frac{dy}{D} = n\lambda + d\theta_0$

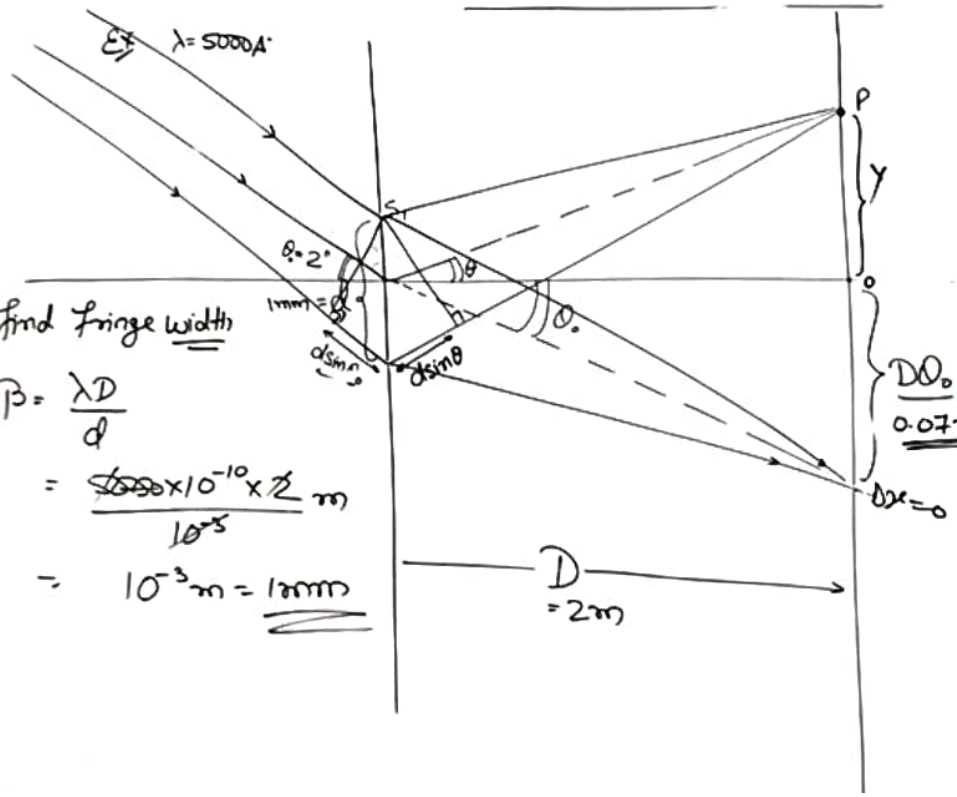
$y = \frac{n\lambda D}{d} + D\theta_0$

Central maxima $n = 0$ $y = D\theta_0$

1st maxima $\Rightarrow n = 1$ $y = \frac{\lambda D}{d} + D\theta_0$

$= \beta + D\theta_0$

Oblique incidence of light / changing dirn of incident light



find fringe width

$\beta = \frac{\lambda D}{d}$

$= \frac{5000 \times 10^{-10} \times 2}{10^{-3}}$ m

$= 10^{-3}$ m = 1mm

find position of maximas

$Y = -0.07m \pm n\beta$

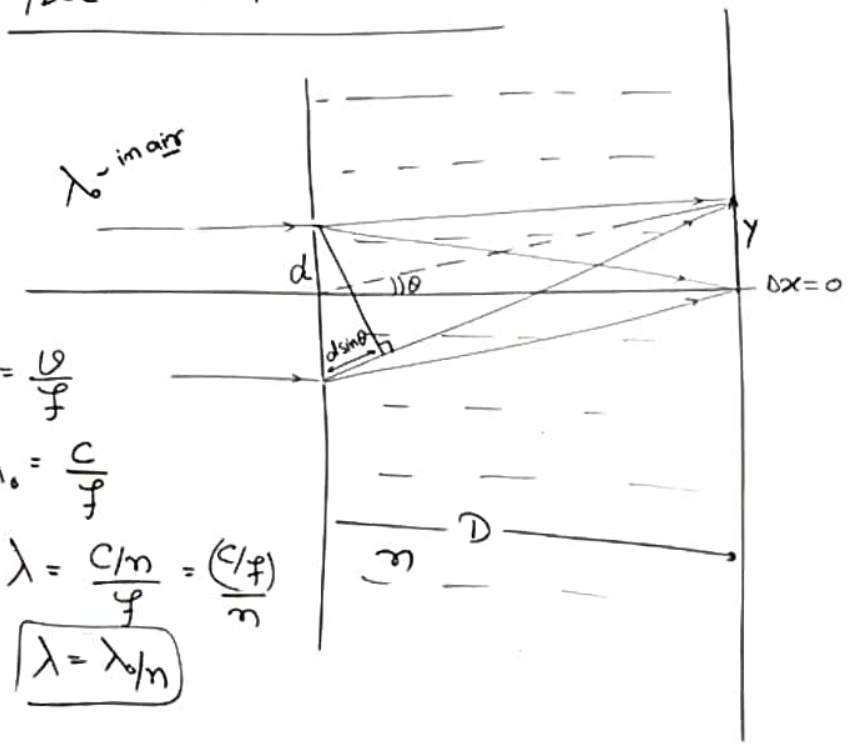
$= (0.07 \pm n \times 10^{-3})$ meter

find position of minimas

$Y = -0.07m \pm (n - \frac{1}{2})\beta$

$= \{0.07 \pm (n - \frac{1}{2}) \times 10^{-3}\}$ meter

YDSE in transparent medium



$\lambda = \frac{v}{f}$

$\lambda_0 = \frac{c}{f}$

$\lambda = \frac{c/n}{f} = \frac{(c/f)}{n}$

$\boxed{\lambda = \lambda_0/n}$

$\beta_0 = \frac{\lambda_0 D}{d}$

$y = n\beta_0$

$y = n\beta$

pattern shrink

$\Delta x = d\sin\theta$
$= d\tan\theta$
$= \frac{dy}{D}$

maxima'

$\Delta x = n\lambda$

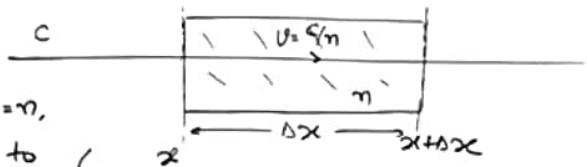
$\frac{dy}{D} = n\lambda$

$y = n\frac{\lambda D}{d}$

$\boxed{\beta = \frac{\lambda D}{d} = \frac{\lambda_0 D}{n d} = \frac{\beta_0}{n}}$

Optical length

travelling Δx distance in medium of R.I. $= n$, is equivalent to travel "$n\Delta x$" distance in vacuum / air.



$n\Delta x$ – optical length

Δx = geometrical length.

$E = E_0 \sin(\omega t - kx)$

$= E_0 \sin\omega\left(t - \frac{x}{v}\right)$

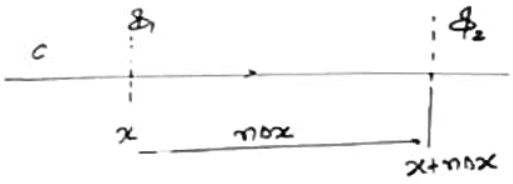
ϕ

$\phi_1 = \omega\left\{t - \frac{x}{v}\right\}$

$\phi_2 = \omega\left\{t - \frac{x+\Delta x}{v}\right\}$

phase difference

$= \omega\frac{\Delta x}{v} = \frac{\omega(n\Delta x)}{c}$ ✓



$E = E_0 \sin\omega\left(t - \frac{x}{c}\right)$

$\phi_1 = \omega\left(t - \frac{x}{c}\right)$

$\phi_2 = \omega\left(t - \frac{x+n\Delta x}{c}\right)$

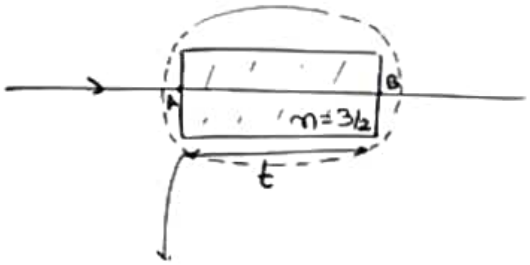
$\boxed{\Delta\phi = \frac{\omega n\Delta x}{c}}$ ✓

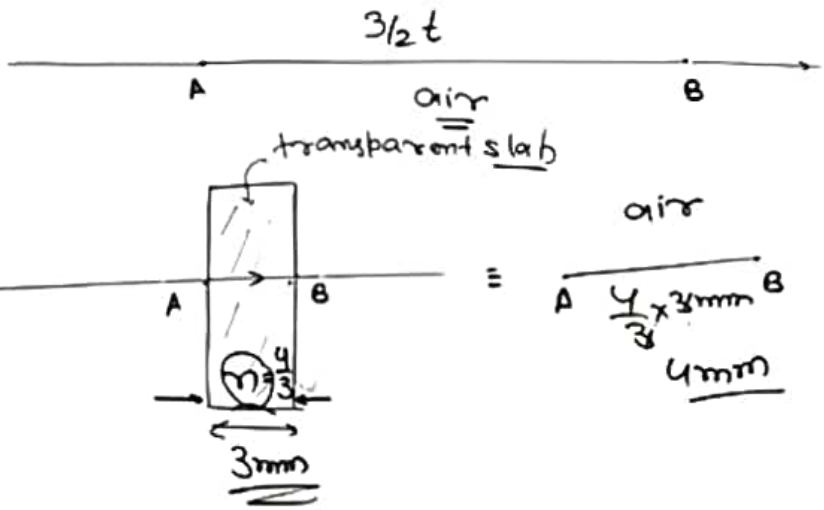
Optical length

travelling Δx — $\frac{c}{}$

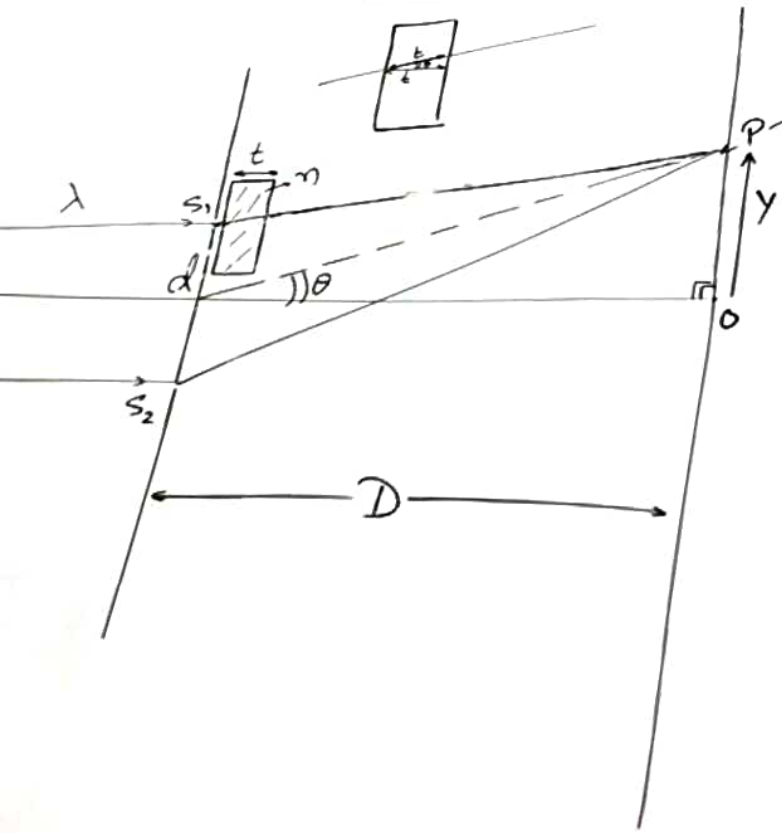
distance in medium of R.I. $= n$, is equivalent to travel "$n\Delta x$" distance in vacuum / air. $\Delta x =$

$n\Delta x$ – optical length





Insertion of parallel transparent slab



path difference

$$\Delta x = S_2P - S_1P'$$

$$= S_2P - \{S_1P - t + nt\}$$

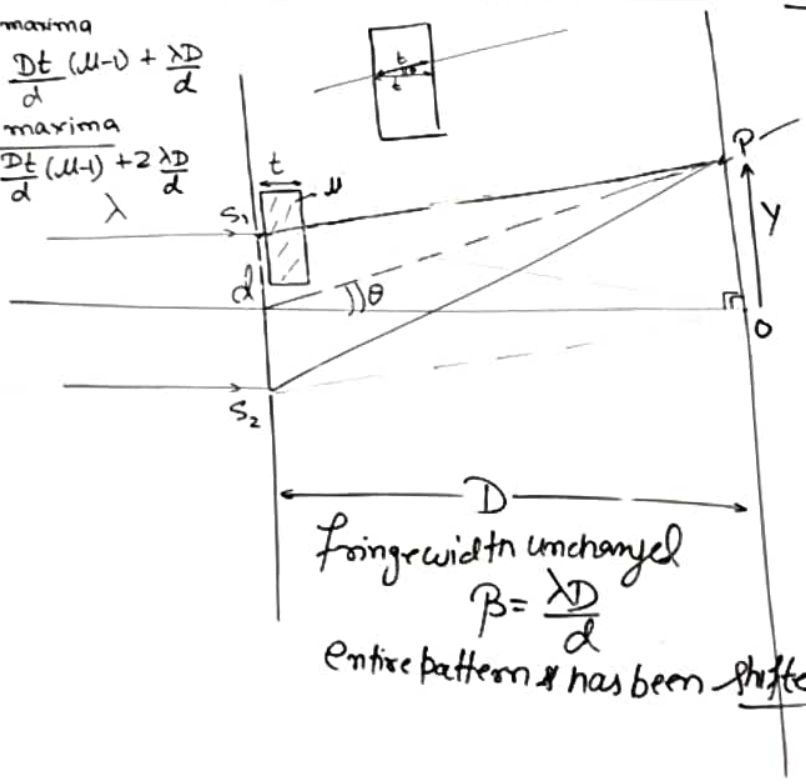
$$= (S_2P - S_1P) - (n-1)t$$

$$= d\sin\theta - (n-1)t$$

$$\approx d\tan\theta - (n-1)t$$

$$\Delta x \approx \frac{dy}{D} - (n-1)t$$

Insertion of parallel transparent Slab



1st maxima

$$y = \frac{Dt}{d}(\mu-1) + \frac{\lambda D}{d}$$

2nd maxima

$$y = \frac{Dt}{d}(\mu-1) + 2\frac{\lambda D}{d}$$

λ

maxima's

Minima's

$$y = \left(n-\frac{1}{2}\right)\beta + \text{shift}$$

$$y = \left(n-\frac{1}{2}\right)\beta + \frac{Dt}{d}(\mu-1)$$

$$y = \frac{Dt}{d}(\mu-1) + n\frac{\lambda D}{d}$$

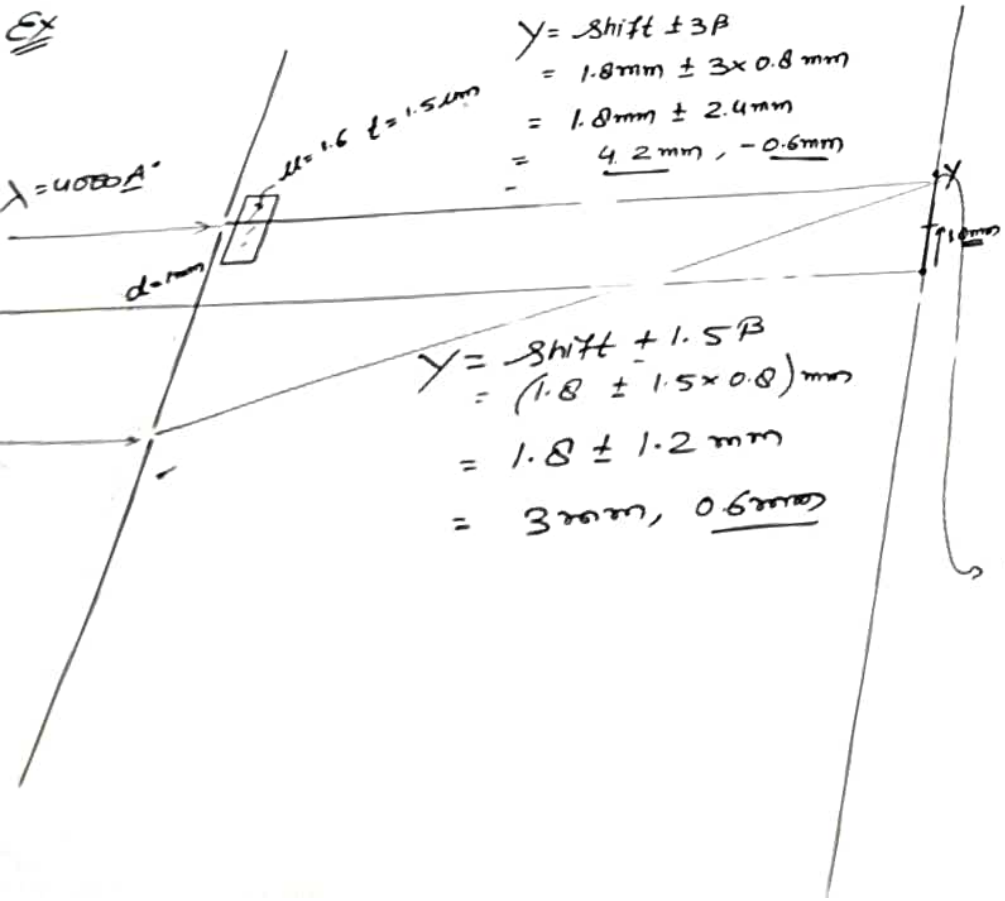
$$= \frac{Dt}{d}(\mu-1) + n\beta$$

fringe width unchanged

$$\beta = \frac{\lambda D}{d}$$

Entire pattern has been shifted by $\frac{Dt}{d}(\mu-1)$

Ex



Y = shift ± 3β
= 1.8mm ± 3 × 0.8 mm
= 1.8mm ± 2.4mm
= 4.2 mm, −0.6mm

Y = shift ± 1.5β
= (1.8 ± 1.5 × 0.8) mm
= 1.8 ± 1.2 mm
= 3mm, 0.6mm

find ① fringe width
② shift in pattern due to thin film.
③ find position of 3rd maxima
④ find position of 2nd minima.

$$\beta = \frac{\lambda D}{d} = \frac{4000 \times 10^{-10} \times 2}{10^{-3}}\ m$$

$$= 0.8\ mm$$

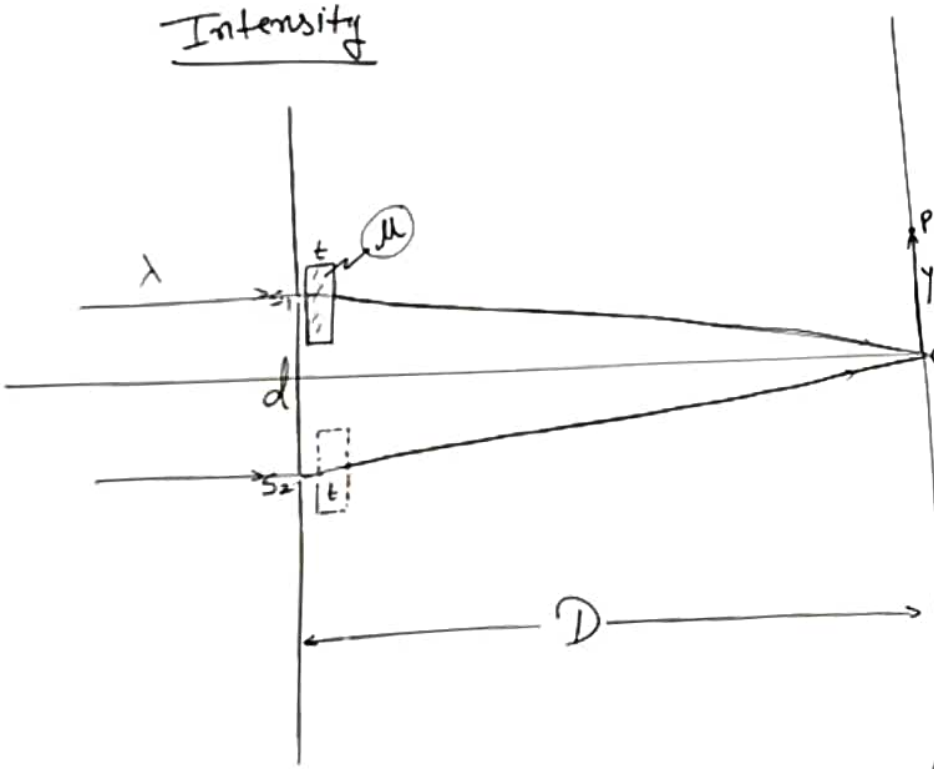
$$\Delta x = S_2P - S_1P'$$

$$= S_2P - \{S_1P - t + \mu t\}$$

$$= S_2P - S_1P - (\mu - 1)t$$

$$0 = \frac{dY}{D} - (\mu - 1)t$$

Intensity



I_0, I_0

phase difference btw waves at point O $= \Delta\phi = \frac{2\pi}{\lambda}\Delta x$

$= \frac{2\pi}{\lambda}(\mu-1)t$

$\Delta x = \mu t - t$

$= (\mu-1)t$

$I_{Res} = 4I_0 \cos^2\frac{\Delta\phi}{2}$

$(I_{Res})_0 = 4I_0 \cos^2\frac{\pi}{\lambda}(\mu-1)t$

I_1, I_2

$I_{Res} = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\frac{2\pi}{\lambda}(\mu-1)t$

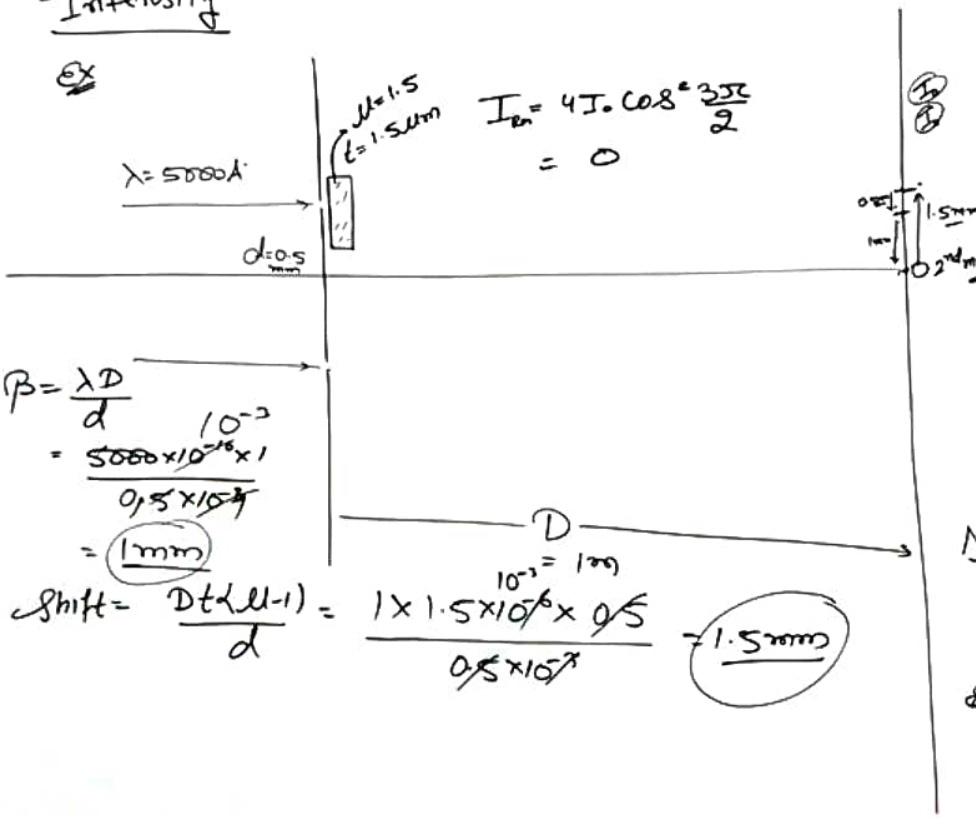
$P \Rightarrow \Delta x = \frac{dy}{D} - (\mu-1)t$

$\Delta\phi = \frac{2\pi}{\lambda}\left\{\frac{dy}{D} - (\mu-1)t\right\}$

$(I_{Res})_P = 4I_0 \cos^2\frac{\pi}{\lambda}\left\{\frac{dy}{D} - (\mu-1)t\right\}$

Intensity

Ex



$I_{net} = 4I_0 \cos^2 \frac{3\pi}{2} = 0$

$\beta = \frac{\lambda D}{d} = \frac{5000 \times 10^{-10} \times 1}{0.5 \times 10^{-3}} = 1\text{ mm}$

$\text{Shift} = \frac{Dt(\mu-1)}{d} = \frac{1 \times 1.5 \times 10^{-6} \times 0.5}{0.5 \times 10^{-3}} = 1.5\text{ mm}$

find

1. β
2. Shift due to thin slab
3. position of nth maxima
4. — — n^{th} minima
5. Intensity at O

n^{th} maxima $y_n = 1.5\text{ mm} \pm n \times 1\text{ mm}$

n^{th} minima $y_n = 1.5\text{ mm} \pm \left(n - \frac{1}{2}\right) \times 1\text{ mm}$

$\Delta x = (\mu-1)t = 0.5 \times 1.5 \times 10^{-6} = 0.75 \times 10^{-6}\text{ m}$

$\Delta\phi = \frac{2\pi}{5000 \times 10^{-10}} \times 0.75 \times 10^{-6} = 3\pi$

Fresnel Biprism

A ~ small small angled prism

$\delta = (\mu - 1)A$

β = fring width $= \frac{\lambda D}{d} = \frac{\lambda(a+b)}{2a\delta}$

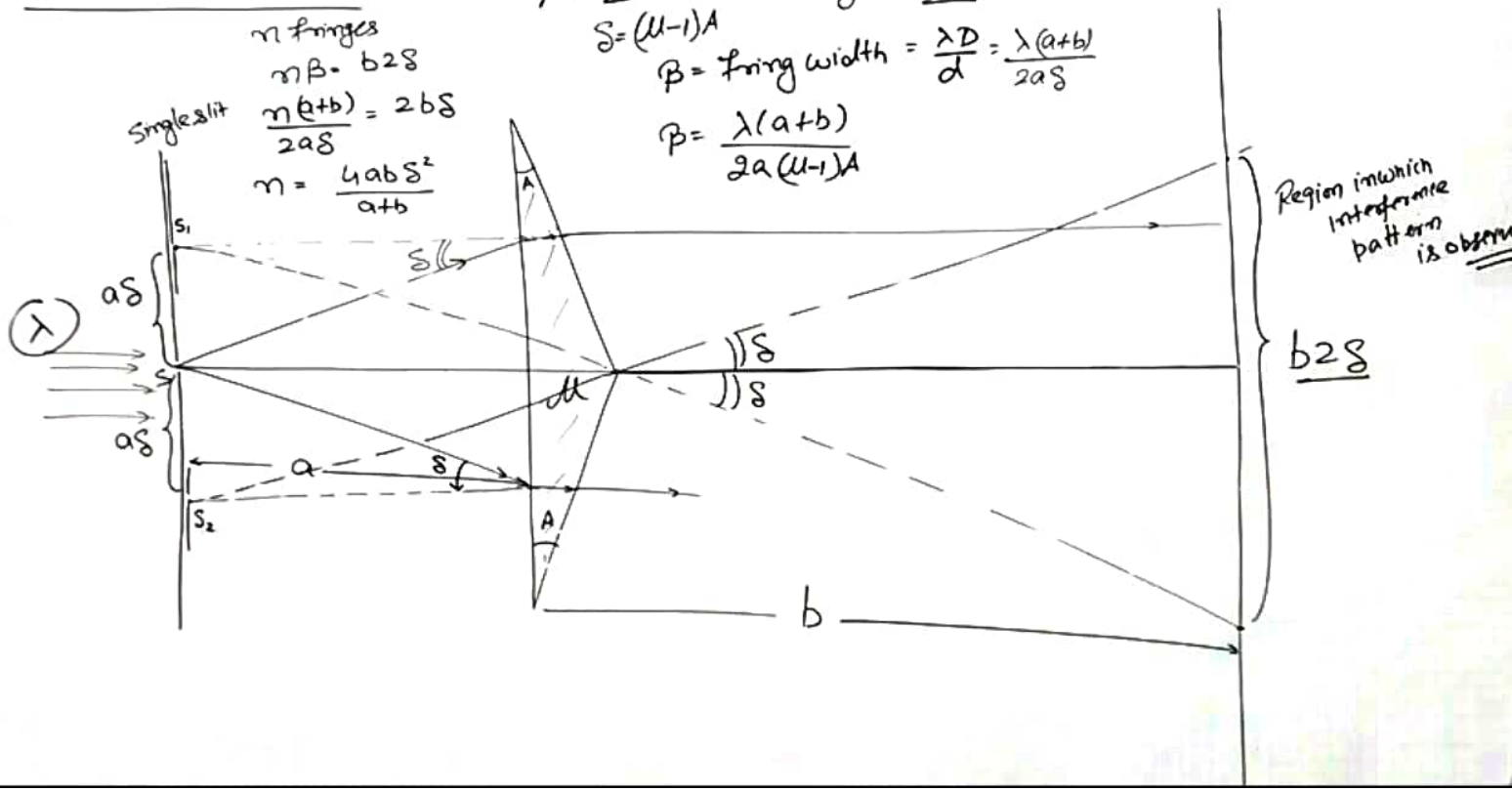
$\beta = \frac{\lambda(a+b)}{2a(\mu-1)A}$

n fringes

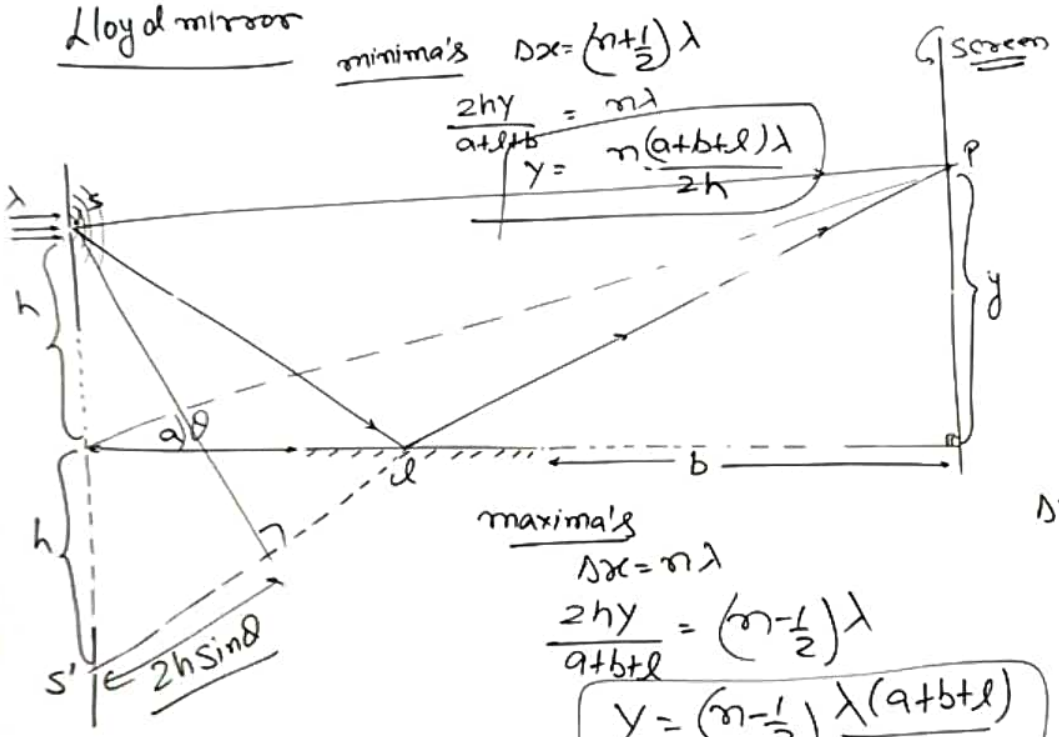
$n\beta = b2\delta$

$\frac{n(a+b)}{2a\delta} = 2b\delta$

$n = \frac{4ab\delta^2}{a+b}$



Lloyd mirror



minima's $\Delta x = \left(n+\frac{1}{2}\right)\lambda$

$$\frac{2hy}{a+l+b} = n\lambda$$

$$y = \frac{n(a+b+l)\lambda}{2h}$$

$$\beta = \frac{\lambda D}{d}$$

$$= \frac{\lambda(a+l+b)}{2h}$$

$$\Delta x = (S'P - SP) + \frac{\lambda}{2}$$

$$= 2h\sin\theta + \frac{\lambda}{2}$$

$$= 2h\tan\theta + \frac{\lambda}{2}$$

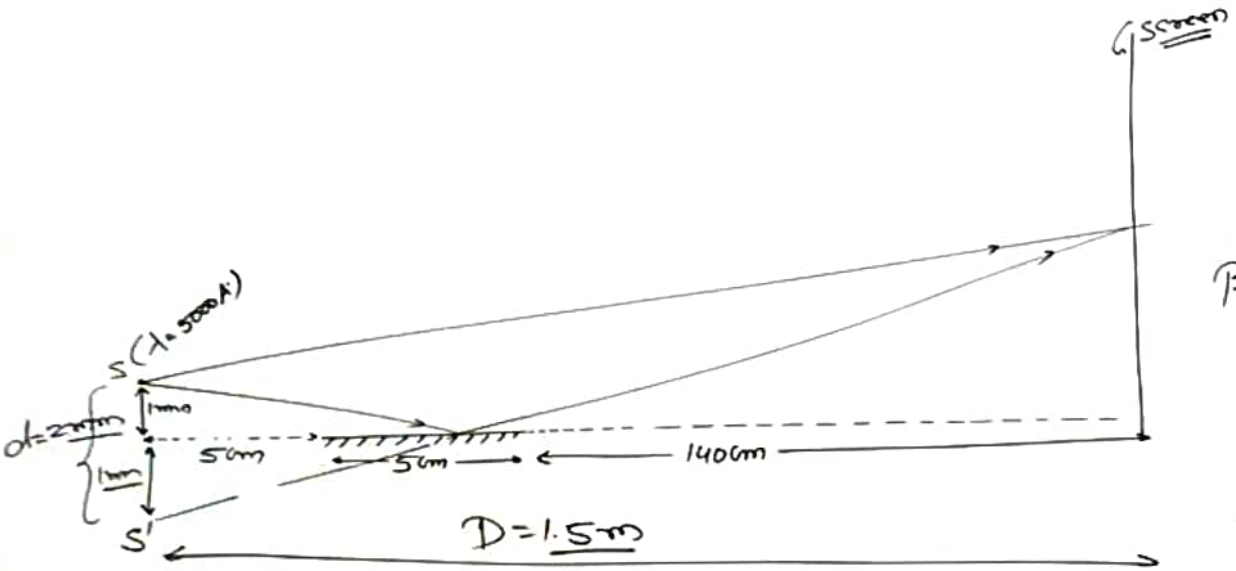
$$\Delta x = \frac{2h.y}{(a+l+b)} + \frac{\lambda}{2}$$

maxima's

$$\Delta x = n\lambda$$

$$\frac{2hy}{a+b+l} = \left(n-\frac{1}{2}\right)\lambda$$

$$y = \left(n-\frac{1}{2}\right)\frac{\lambda(a+b+l)}{2h}$$



Find fringe width of interference pattern on screen

$$\beta = \frac{\lambda D}{d}$$

$$= \frac{5000 \times 10^{-10} \times 1.5}{2 \times 10^{-3}}$$

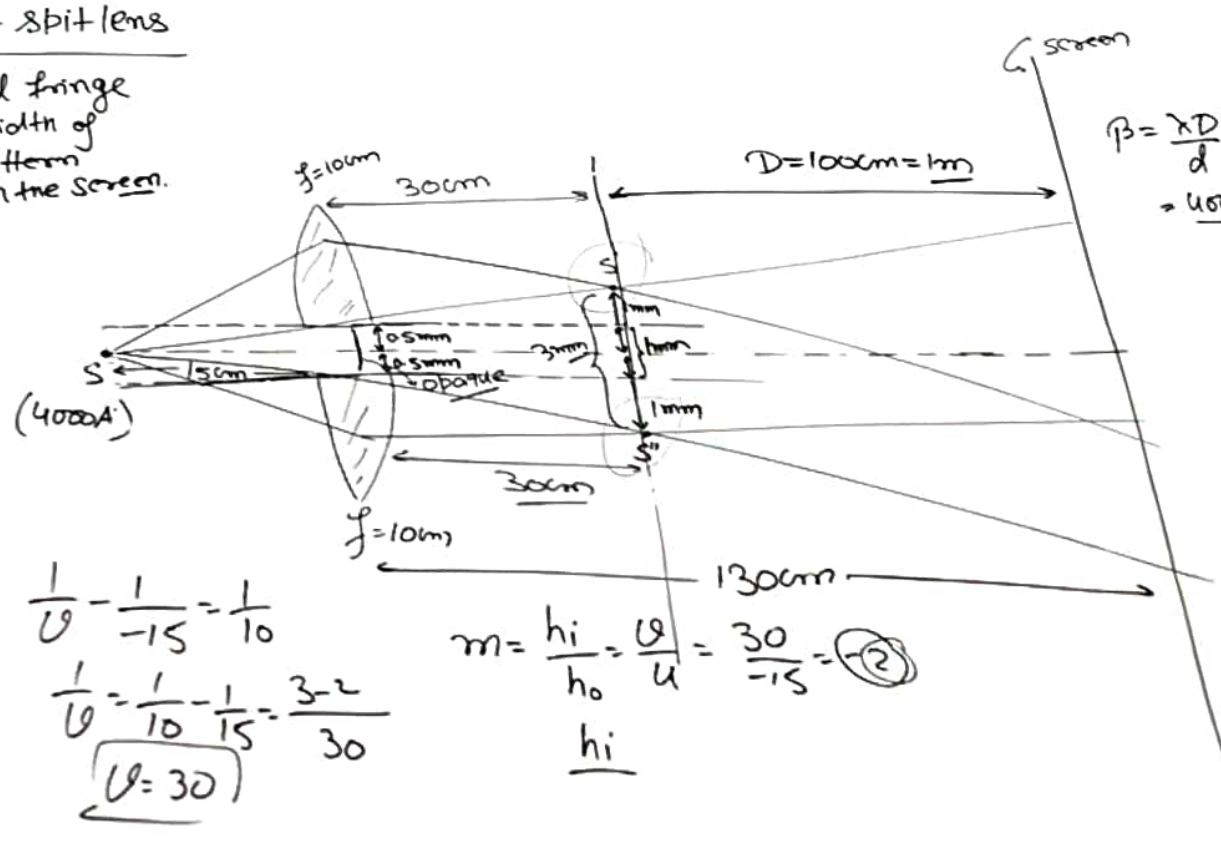
$$= 2.5 \times 1.5 \times 10^{-4} \text{ m}$$

$$= 3.75 \times 10^{-4} \text{ m}$$

$$= 0.375 \text{ mm}$$

Billet Split lens

Ex find fringe width of pattern on the screen.



$$\beta = \frac{\lambda D}{d} = \frac{4000\times10^{-10}\times}{3\times10^{-3}}$$

$$\frac{1}{v} - \frac{1}{-15} = \frac{1}{10}$$

$$\frac{1}{v} = \frac{1}{10} - \frac{1}{15} = \frac{3-2}{30}$$

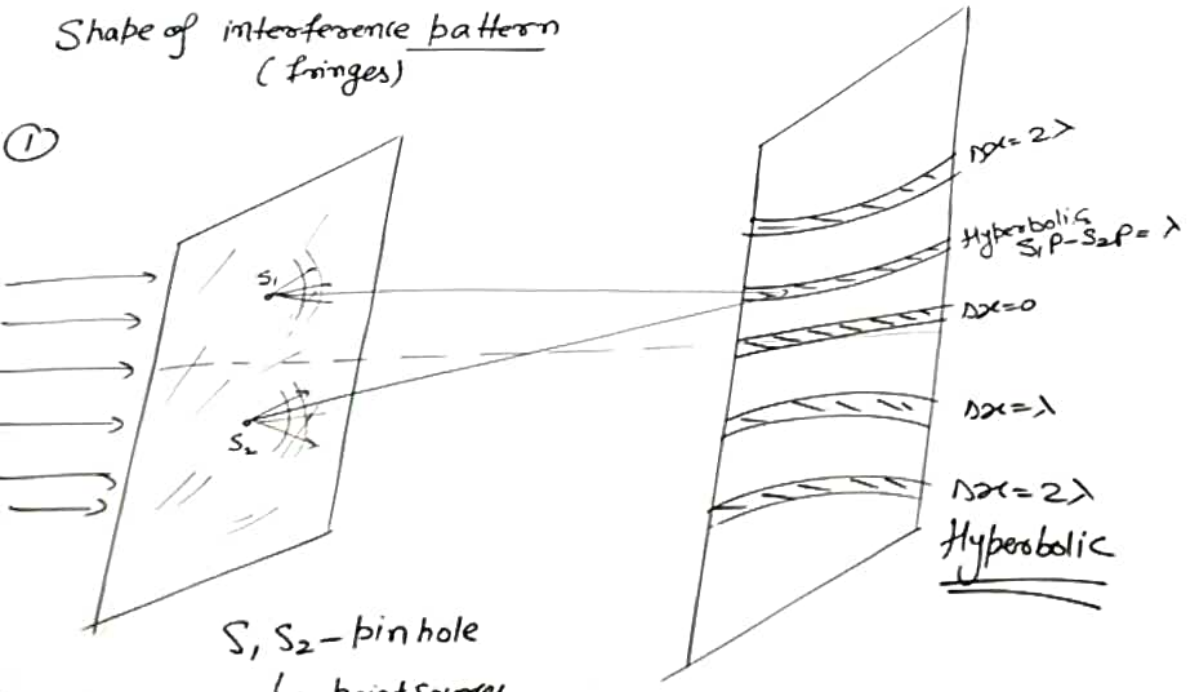
$$\boxed{v = 30}$$

$$m = \frac{h_i}{h_o} = \frac{v}{u} = \frac{30}{-15} = -2$$

hi

Shape of interference pattern
(fringes)

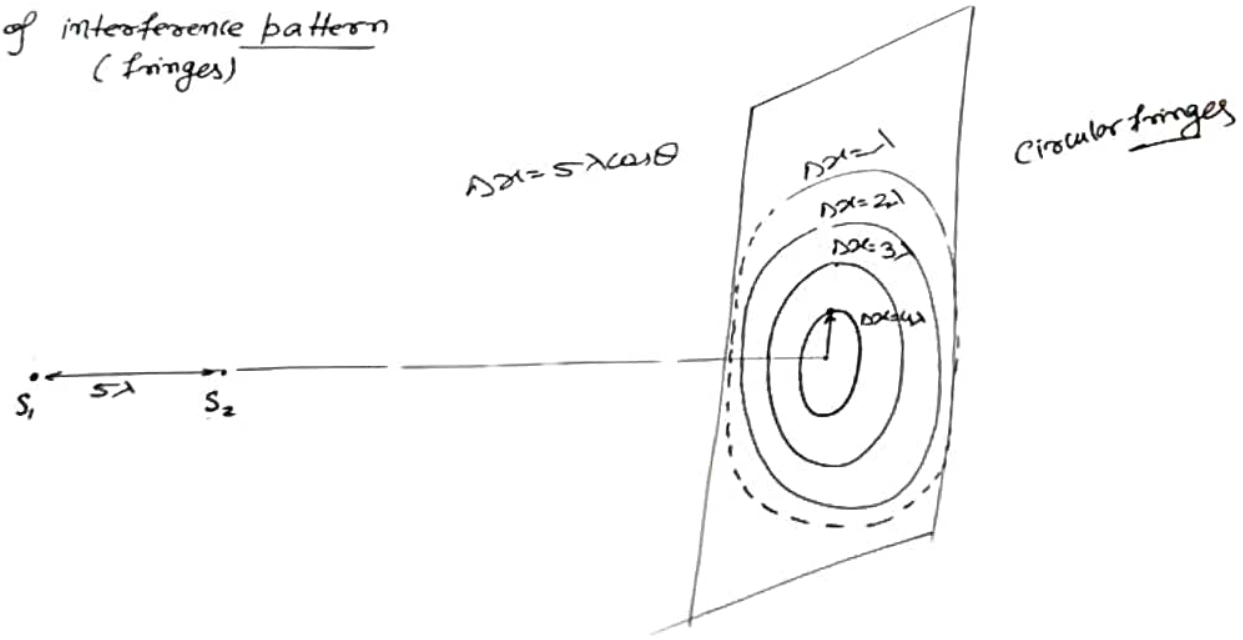
(1)



S_1, S_2 – pin hole
↳ point sources

Shape of interference pattern
(fringes)

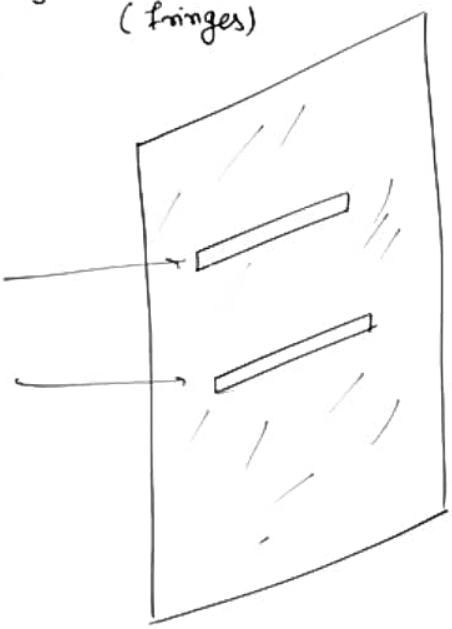
(2)

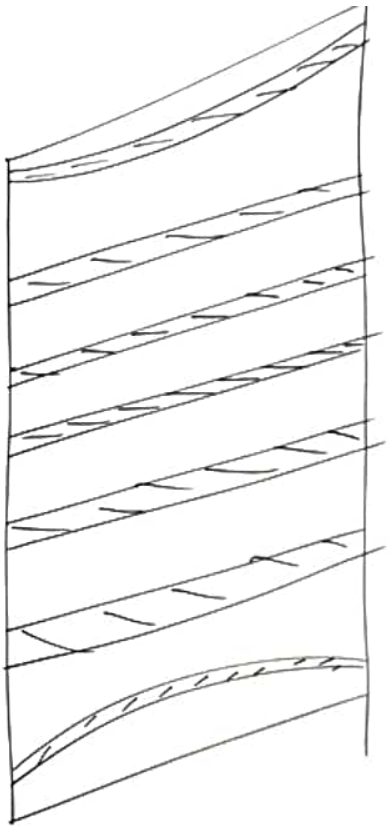


Circular fringes

Shape of interference pattern
(fringes)

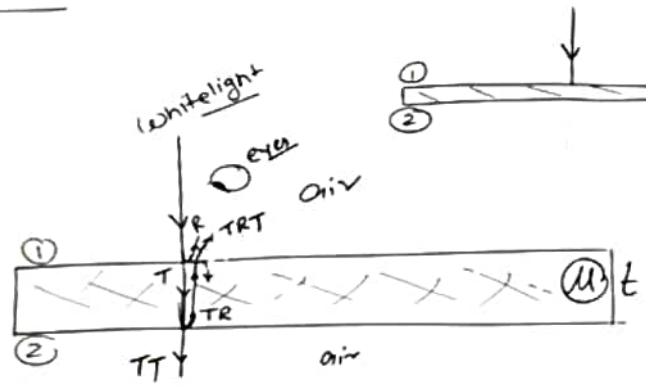
③





Thin film interference

Surface ①





R and TRT : interference - destructively (I_{min})

$R \rightarrow \phi_1 = \pi$

$TRT = \phi_2 = \frac{2\pi}{\lambda}\{2\mu t\}$

phase difference

$\Delta\phi = \phi_2 - \phi_1 = \frac{2\pi}{\lambda}\{2\mu t\} - \pi =$

$2\frac{n}{\lambda}\mu t - \pi = (2n-1)\pi$

$2\mu t = n\lambda$

$\lambda = \frac{2\mu t}{n}$

→ I_{min}

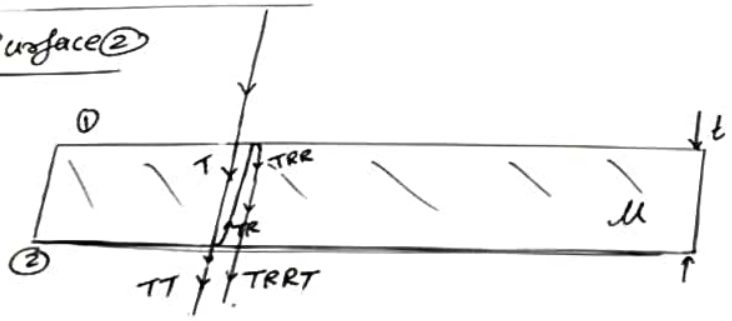
→ Poorly Reflected

→ Dark.

competishun

Thin film interference

Surface ②



observer

$4\mu t = (2n+1)\lambda$

$\lambda = \frac{4\mu t}{2n+1}$ → Poorly transmitted

Surface ② — dark
- Imin
- destructive interference

TT & TRRT ⇒ $\Delta\phi = (2n+1)\pi$

TT ⇒ $\phi_1 = 0$

TRRT ⇒ $\phi_2 = \frac{2\pi}{\lambda}\{2\mu t\}$

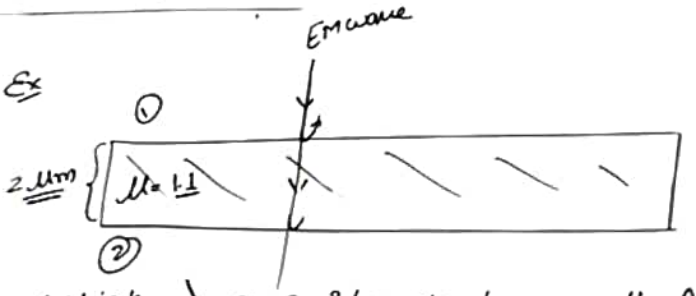
phase difference

$= \Delta\phi = \phi_2 - \phi_1$

$= \frac{2\pi}{\lambda}(2\mu t) = (2n+1)\pi$

Thin film interference

Ex



Which λ are strongly transmitted.

$2\mu t = n\lambda$

$\lambda = \frac{2 \times 1.1 \times 2 \times 10^{-6}}{n}$

$= \frac{4.4 \times 10^{-6}}{n} = \frac{4400}{n}$ nm

$= 4400\,nm, \frac{4400}{2}\,nm, \frac{4400}{3}\,nm,$

---

find λ which are strongly Reflected poorly Refracted transmitted

$2\mu t = (2n+1)\lambda$

$\lambda = \frac{4 \times 1.1 \times 2 \times 10^{-6}}{(2n+1)}$

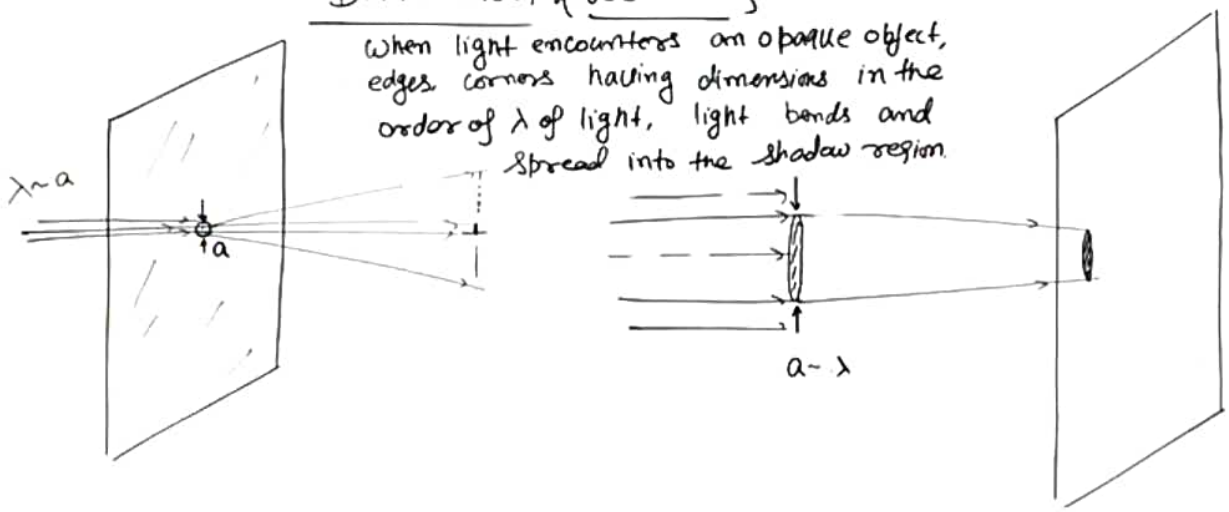
$= \frac{8.8 \times 10^{-6}}{2n+1}$ m

$= \frac{8800}{(2n+1)}$ nm

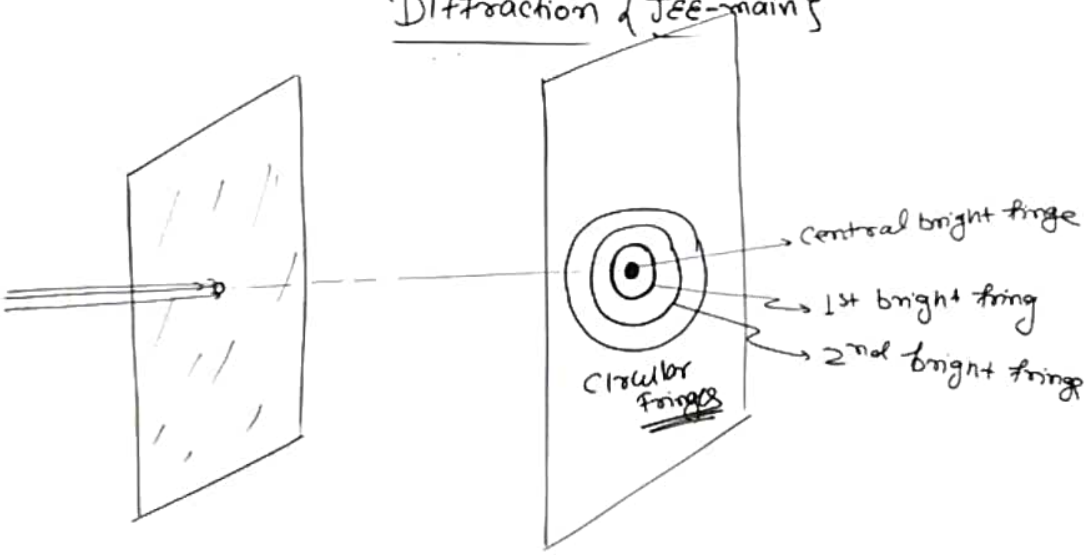
$= \frac{8800}{1}\,nm, \frac{8800}{3}\,nm, \frac{8800}{5}\,nm, \ldots$

Diffraction {JEE-main}

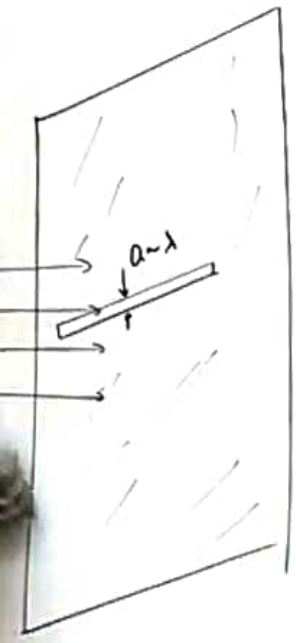
When light encounters an opaque object, edges, corners having dimensions in the order of λ of light, light bends and spread into the shadow region

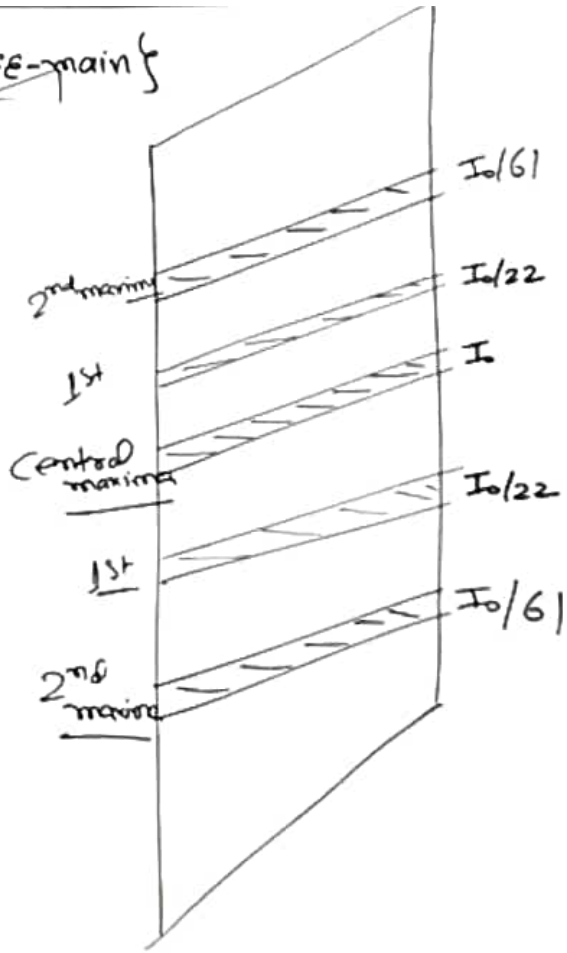


Diffraction {JEE-main}



Diffraction {JEE-main}



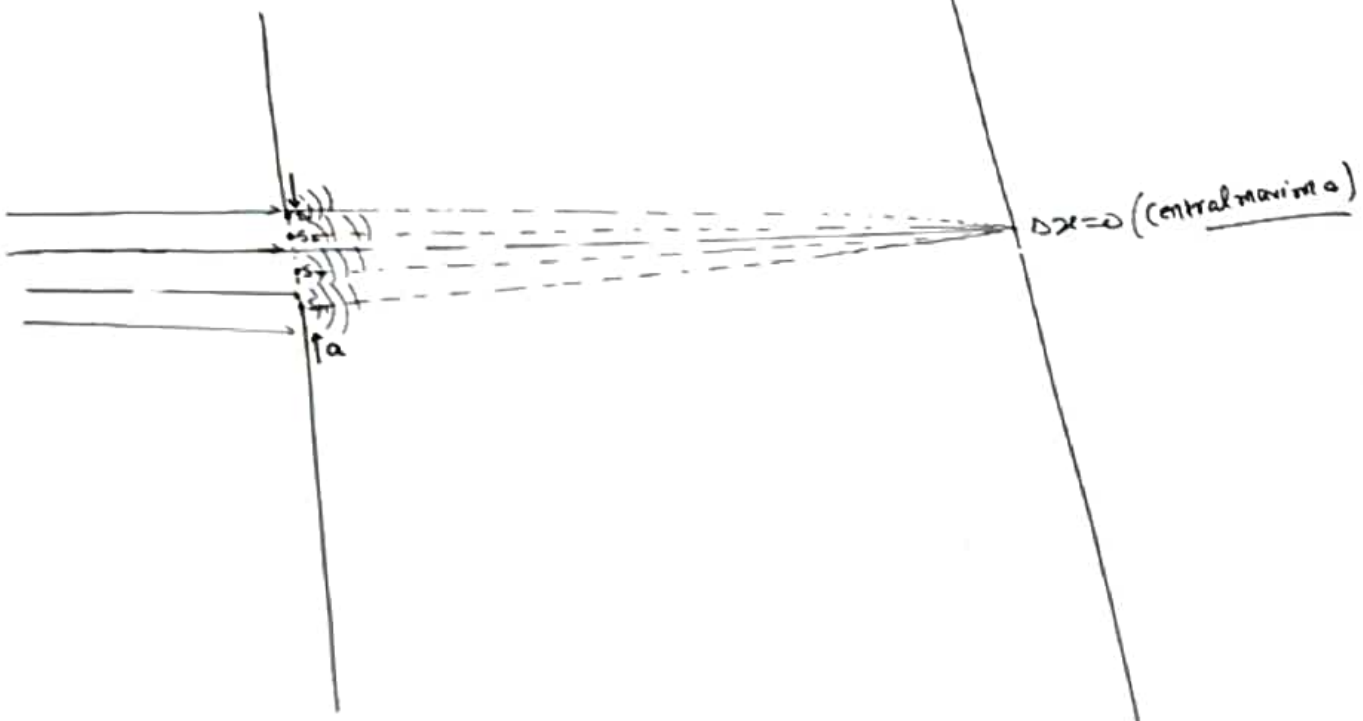


Diffraction {JEE-main}

Single slit diffraction

a – thickness of slit

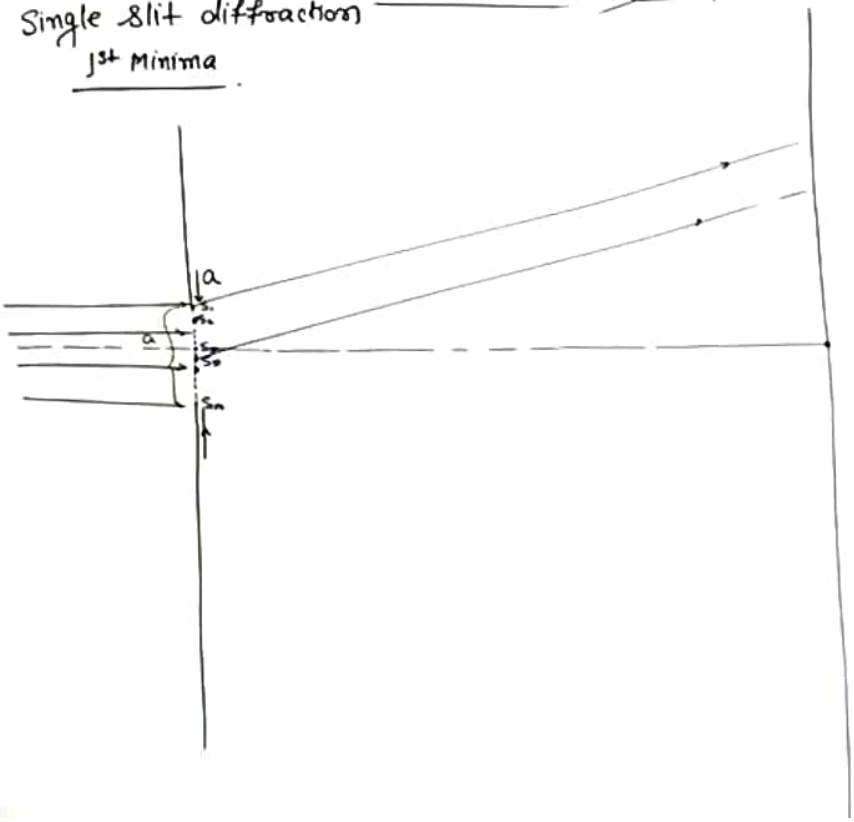
Central maxima:

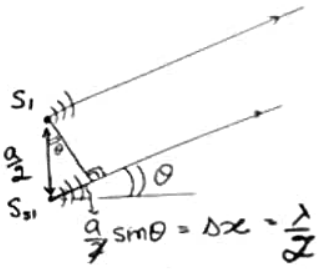


Diffraction {JEE-main}

Single slit diffraction

1st Minima





$\frac{a}{2}\sin\theta = \Delta x = \frac{\lambda}{2}$

$a\sin\theta = \lambda$

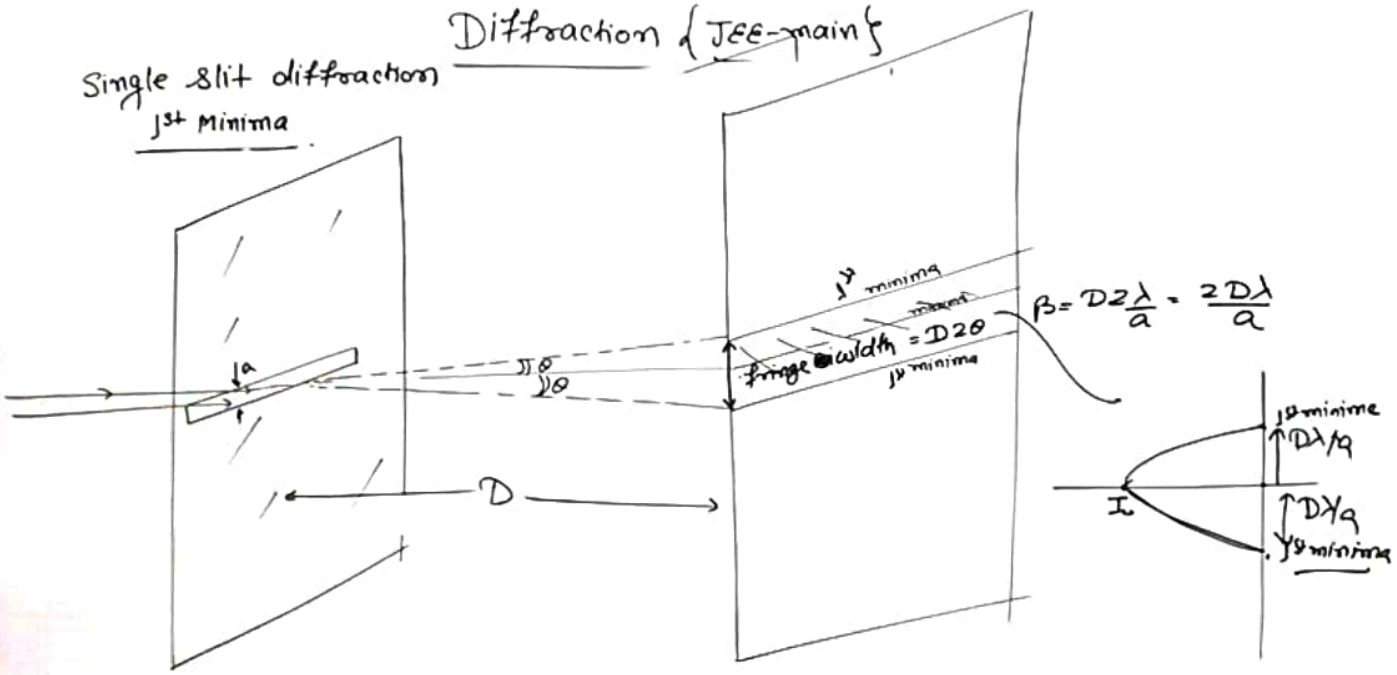
$\sin\theta = \frac{\lambda}{a}$

$\theta = \lambda/a$

Diffraction {JEE-main}

Single slit diffraction

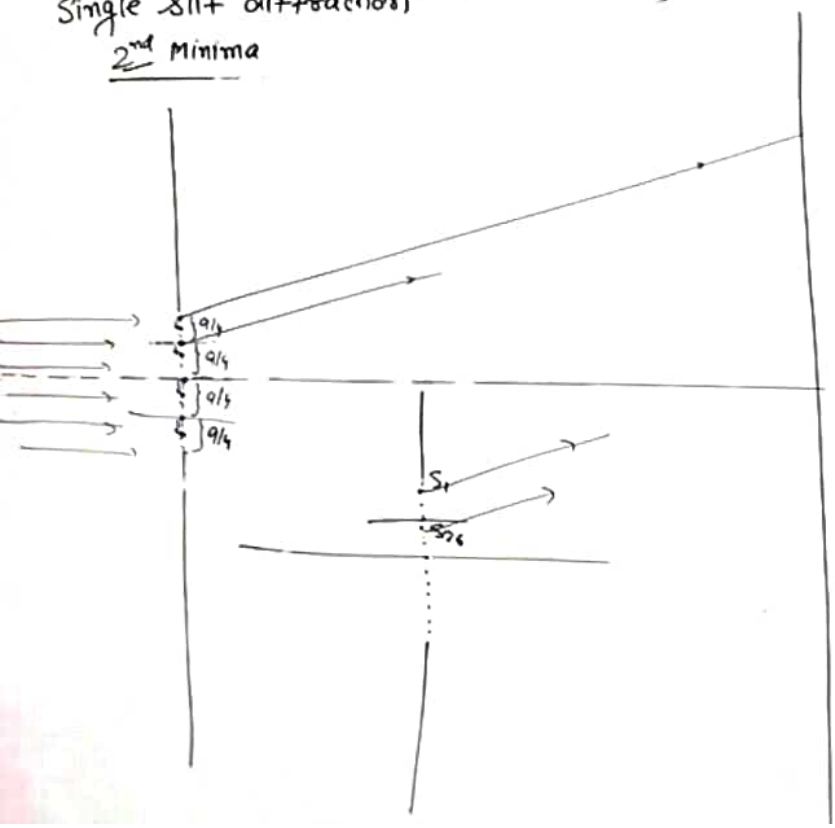
1st Minima



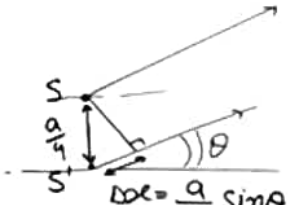
Diffraction (JEE-main)

Single slit diffraction

2nd Minima



Minima $\boxed{\sin\theta_n = \frac{n\lambda}{a}}$ Rem!



$\Delta x = \frac{a}{4}\sin\theta = \frac{\lambda}{2}$

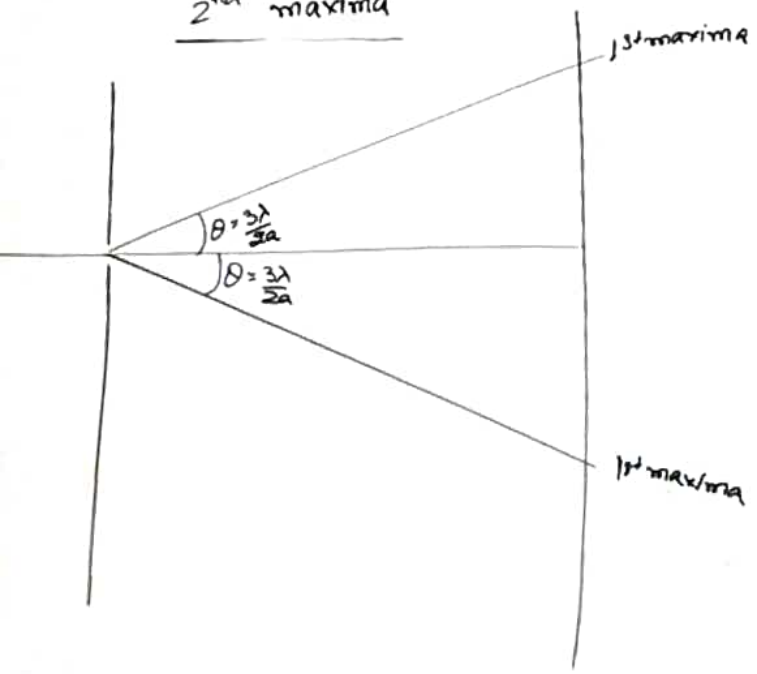
$\sin\theta = \frac{2\lambda}{a}$

$\boxed{\theta \approx \frac{2\lambda}{a}}$ 2nd minima

Diffraction (JEE-main)

Maxima's

2nd maxima

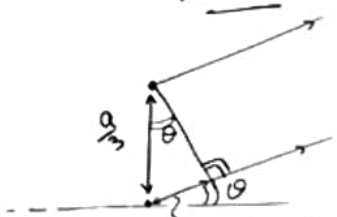


Minima $\sin\theta_n = \frac{n\lambda}{a}$ Rem!

maxima's $\sin\theta_n = \frac{(2n+1)\lambda}{2a}$

$n \neq 0$

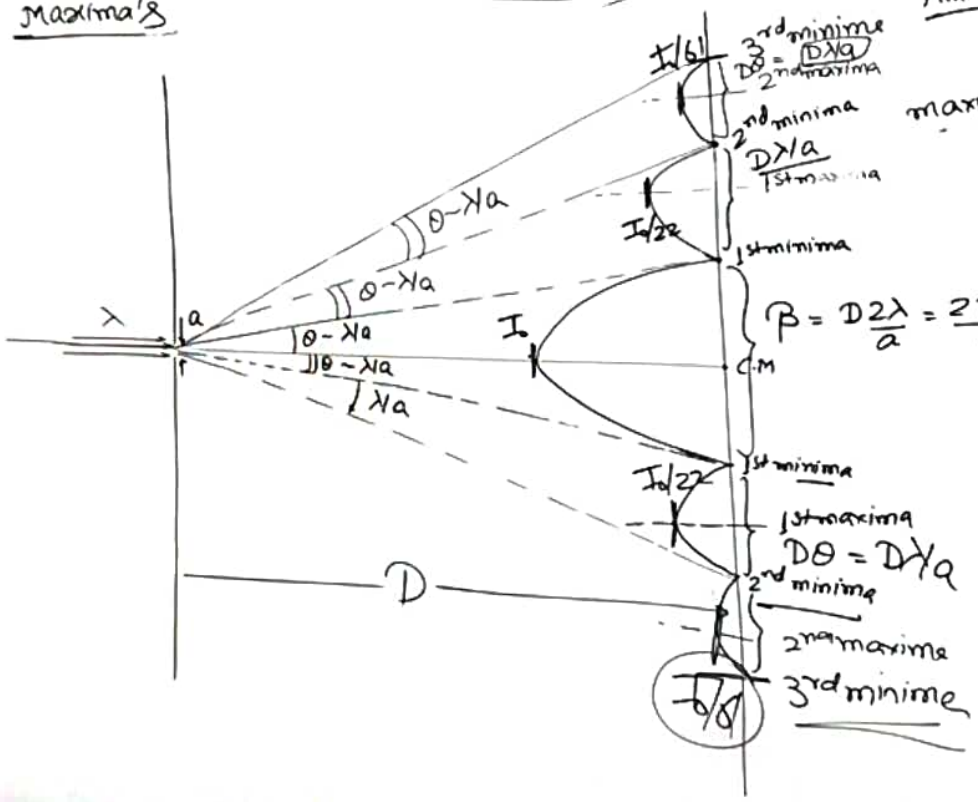
$n = 1, 2, 3, 4$



$\Delta x = \frac{a}{3}\sin\theta = \frac{\lambda}{2}$

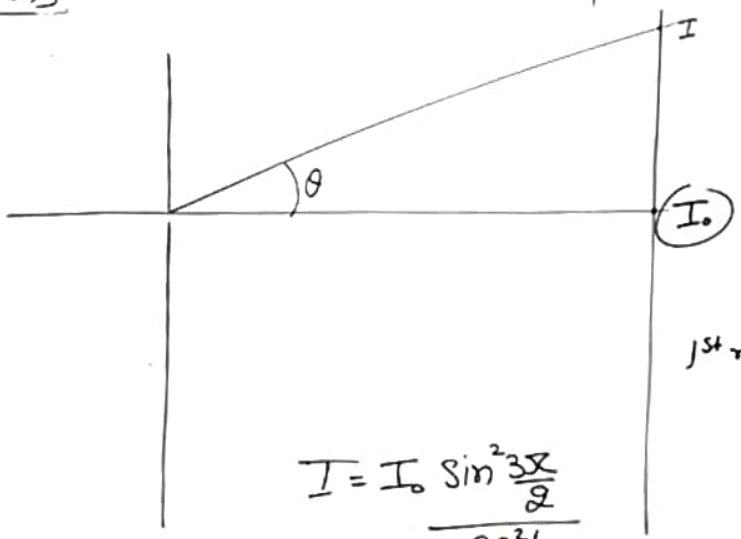
$\sin\theta = \frac{3\lambda}{2a}$

Diffraction (JEE-main)



Diffraction (JEE-main)

Maxima's



Minima $\sin\theta_n = \frac{n\lambda}{a}$ Rem!

$I = I_0 \frac{\sin^2\beta}{\beta^2}$

$\beta = \frac{\pi a \sin\theta}{\lambda}$

1st maxima $\sin\theta = \frac{3\lambda}{2a}$

$\beta = \frac{\pi a \frac{3\lambda}{2a}}{\lambda} = \frac{3\pi}{2}$

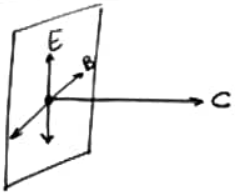
$I = I_0 \frac{\sin^2\frac{3\pi}{2}}{9\pi^2/4}$

$= \frac{4I_0}{9\pi^2} \approx \frac{I_0}{22}$

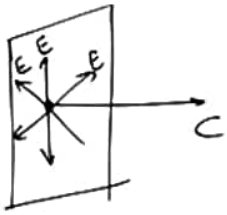
Polarisation of light

(Polarization)

EM wave

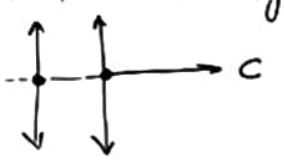


Unpolarised



electric field will oscillate in the entire plan ⊥ to $\vec{c}$ ⇒ unpolarised light

Diagramatically



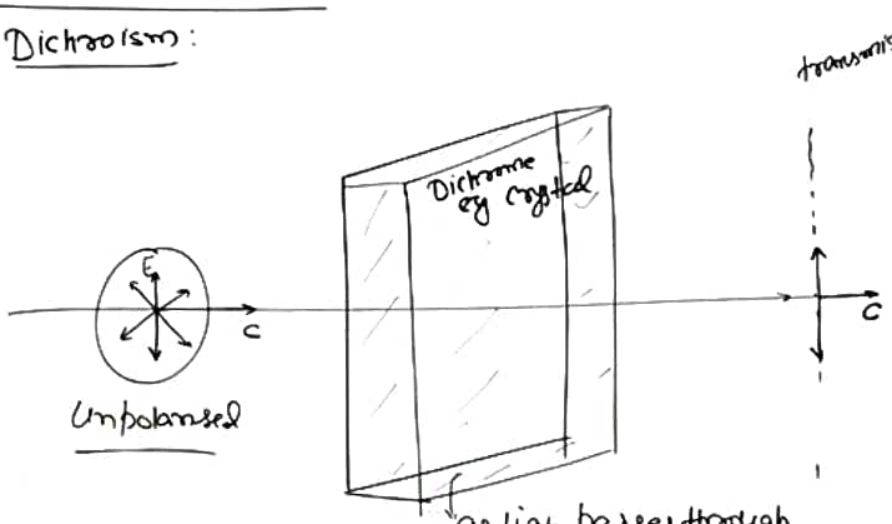
Polarised light:



In polarised light, electric field will oscillate along a fixed line.

How to ~~pt~~ polarise:-

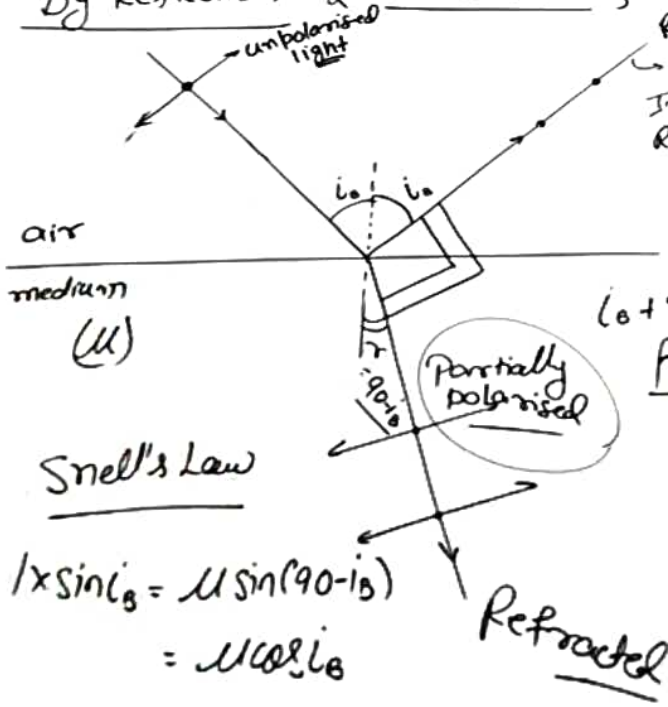
Dichroism:



as ligh passes through this crystal, one component of E.f. will ~~be~~ get absorbed, and one component will be permitted.

In polarised light, electric field will oscillate along a fixed line.

By Reflection { Brewster Law }



In this case, Refracted ray is partially polarised

$i_B + 90 + r = 180$

$r = 90 - i_B$

Snell's Law

$1 \times \sin i_B = \mu \sin(90 - i_B)$

$= \mu \cos i_B$

$\frac{\sin i_B}{\cos i_B} = \mu$

$\tan i_B = \mu$

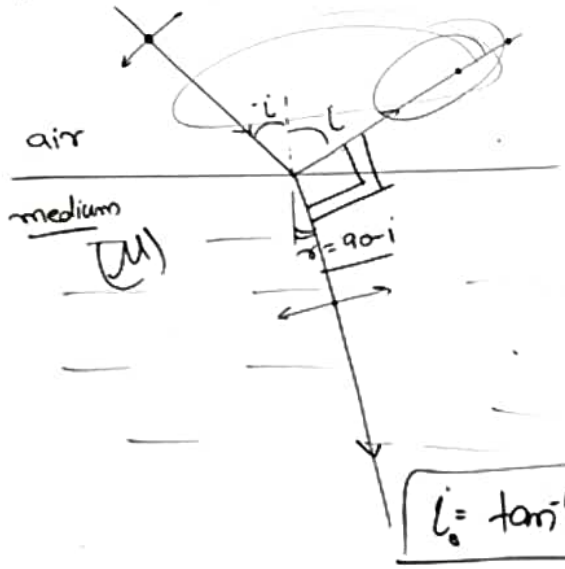
If reflected and refracted ray are ⊥ to each other, then reflected ray is completely polarised.

it happens at specific angle of incidence known as brewster angle (i_B)

$i_B = \tan^{-1}\{\mu\}$ Rem!

In polarised light, electric field will oscillate along a fixed line.

Ex



find 'i', such that reflected ray is completely polarised.

For this arrangement, critical angle 30°.

$\tan i = \mu$

$\sin C = \frac{n_R}{n_D} = \frac{1}{\mu} = \sin 30^\circ = \frac{1}{2}$

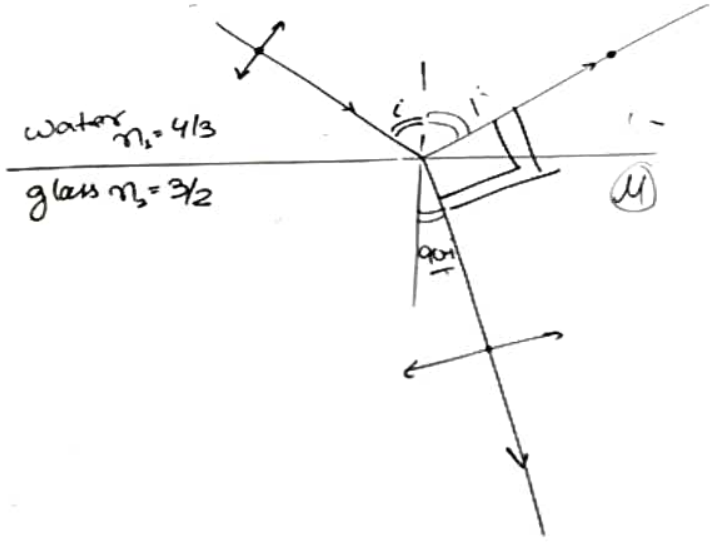
$\mu = 2$

$\tan i = 2$

$i_0 = \tan^{-1}(2)$

In polarised light, electric field will oscillate along a fixed line.

Ex



find 'i', such that reflected ray is completely polarised.

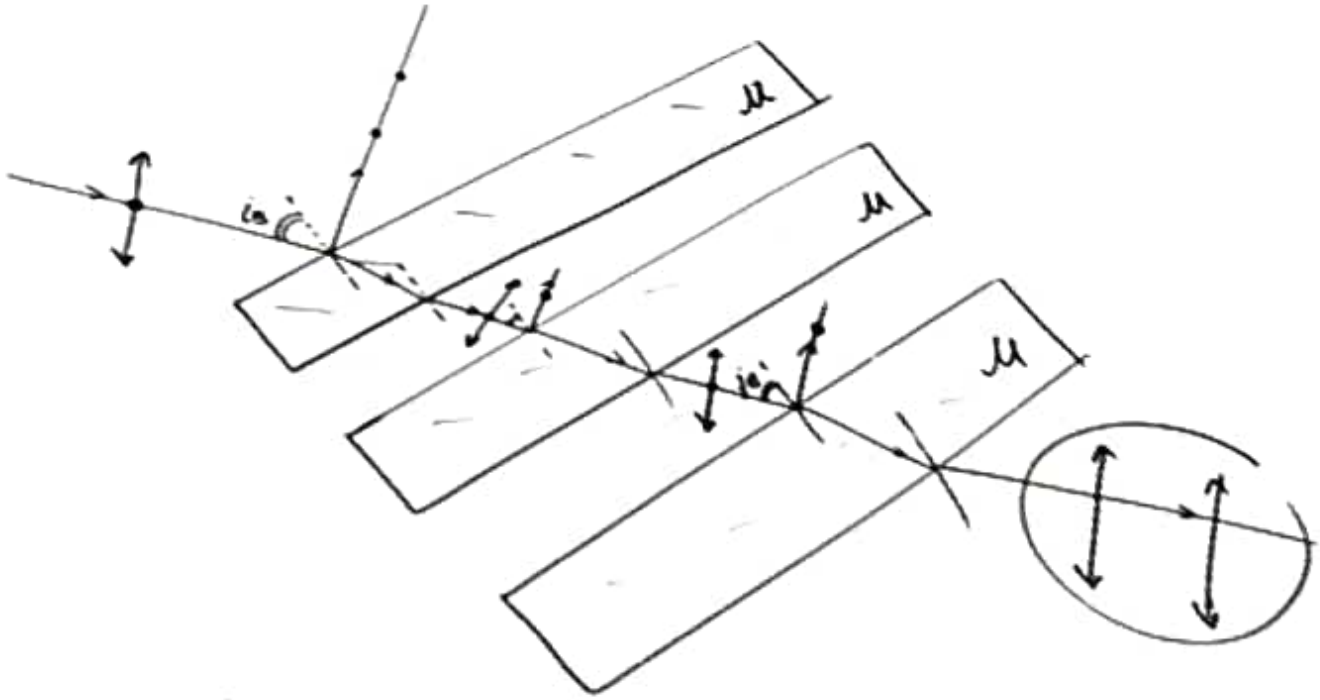
Snell's law

$n_1 \sin i = n_2 \sin(90-i) = n_2 \cos i$

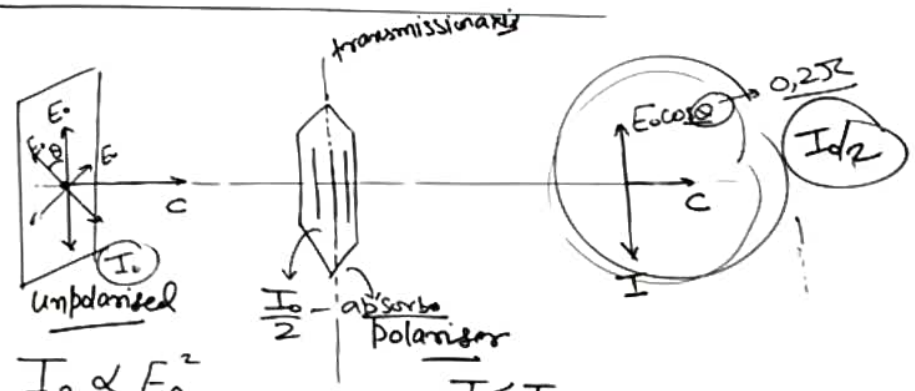
$\tan i = \frac{n_2}{n_1} = \frac{3/2}{4/3} = \frac{9}{8}$

$i = \tan^{-1}\frac{9}{8}$

Polarisation by Refraction



Variation of intensity on polarisation



$I_0 \propto E_0^2$

$I_0 = kE_0^2$

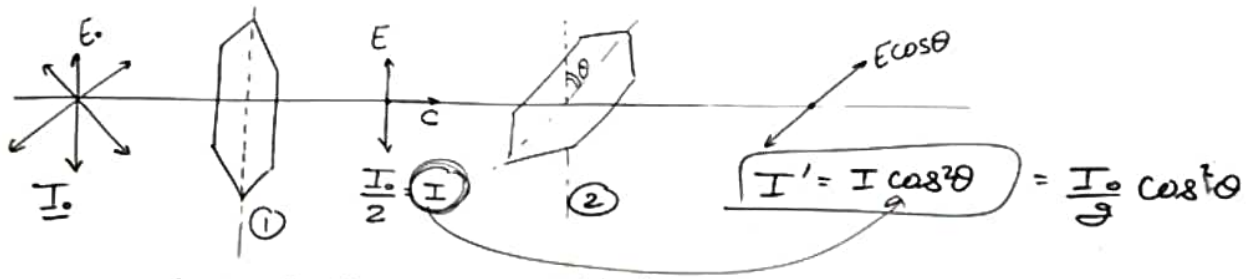
$I < I_0$

$I = kE_0^2\cos^2\theta$ → Avg $= \frac{1}{2}$

$= \frac{kE_0^2}{2} = \frac{I_0}{2}$

if unpolarised light is passed through polarisor, half of the intensity will be absorbed by polariser and only half will be transmitted

Malu's Law

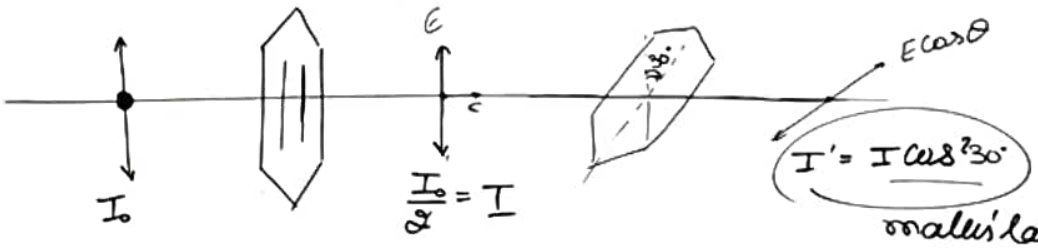


θ - angle blw these 2 polarisers.

$$I = KE^2$$

$$I' = K(E\cos\theta)^2 = KE^2\cos^2\theta$$

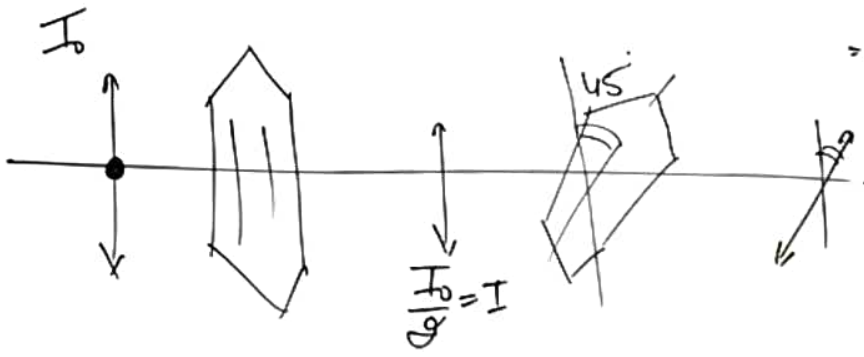
Malus Law

① 

malus law

$= I\left(\frac{3}{4}\right) = \frac{3}{4} I$

$= \frac{3}{8} I_0$

② 

$= I/2$

$= I_0/2$